1 Introductory Activity

Example 1.1. Suppose you're interested in estimating *the proportion of his lifetime that Professor Ross has lived in California.* We'll call this unknown proportion θ (the Greek letter "theta").

(1) If you had to guess a single value for θ, what would your guess be? Why?

Rather than guessing just a single value, we'll assign relative "plausibilities" to a range of values. To illustrate ideas, we'll start by considering only the values 0.2, 0.4, 0.6, 0.8 as possible for θ. (Of course this is unrealistic; we'll consider a more realistic situation soon.) That is, we want to assess our relative plausibility of the statements:

- Professor Ross has lived in CA for 20% of his lifetime
- Professor Ross has lived in CA for 40% of his lifetime
- Professor Ross has lived in CA for 60% of his lifetime
- Professor Ross has lived in CA for 80% of his lifetime

(2) Which of these values — 0.2, 0.4, 0.6, 0.8 — do you think is the *least* plausible for θ? (In other words, which of the four statements in the bullets above do you think is least plausible?)

(3) Which value — 0.2, 0.4, 0.6, 0.8 — do you think is the next least plausible? Roughly, how many *times* more plausible do you think it is than the one from the previous part?

For example, suppose you think that 0.2 is your least plausible and 0.4 is your next least plausible value of θ. If you think that 0.4 is 3 times more plausible than 0.2, then, roughly, you would be willing to engage in a bet that θ is 0.2 versus 0.4 at 3-to-1 odds.

(4) Continue in the manner of the previous part to assign a relative plausibility to each of the values 0.2, 0.4, 0.6, 0.8. Suppose that your least plausible value represents 1 "unit of plausibility". How many "units of plausibility" does each of the other values represent? (Assume that each of the four values has some plausibility, no matter how small; don't assign 0 plausibility to any of the values.)

(5) If 0.2, 0.4, 0.6, 0.8 are the only possible values of θ, then they should account for 100% of your total plausibility. Using your relative plausibilities from the previous part, what percent of your total plausibility does each of the values 0.2, 0.4, 0.6, 0.8 represent? Make a table and a plot of your assessment. Then compare and discuss with someone else in the class.

Example 1.2. The previous example was unrealistic because we only considered the values 0.2, 0.4, 0.6, and 0.8 as plausible. We'll now consider a broader range of values. Since θ is a proportion, we know it must be between 0 and 1, but some values between 0 and 1 might be more plausible to you than others. You'll now sketch a curve over the interval [0, 1] to represent your assessment of relative plausibility. The following questions will help you determine what your sketch should look like.

1. Is it more plausible to you that θ is greater than 0.6 or less than 0.6? (In other words, it is more plausible to you that Professor Ross has lived in CA for more than 60% of his lifetime, or for less than 60% of his lifetime?) How much more plausible?

2. Repeat the previous part for a few values other than 0.6.

3. Fill in the blank: it is equally plausible that θ is greater than or less than [blank].

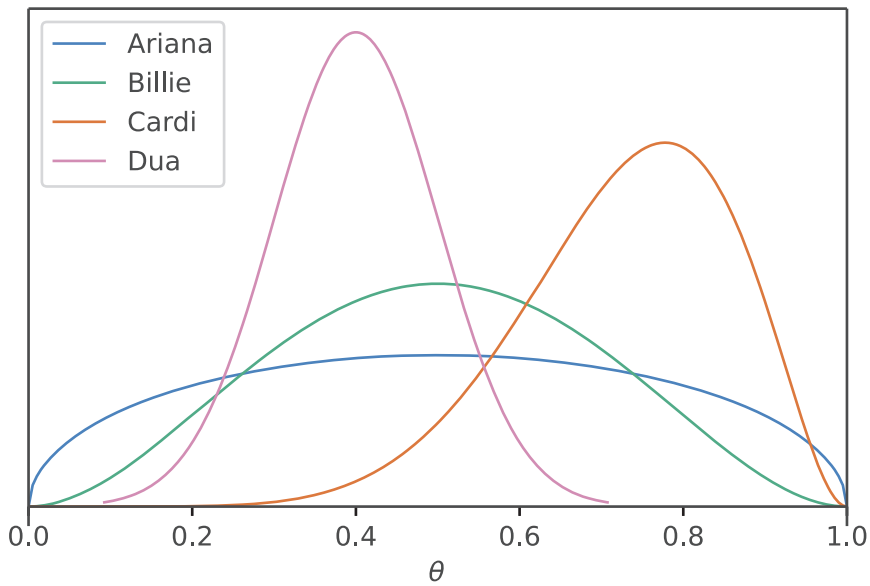
4. Fill in the blank: it is 2 times more plausible that θ is greater than [blank] than less than [blank].

5. Repeat the previous part for a few other ratios (e.g, 3 times).

6. What range of values accounts for most, say 98%, of your plausibility?

7. After your have considered the above questions, sketch a curve that reflects your assesment of plausibility. The plot doesn't have to perfect, but try to reflect your answers to the previous questions as best as you can. (If you get stuck on drawing a sketch, it might help to try Example 1.3 first.)

Example 1.3. The plot below displays the plausibility sketches of four students.



1. Rank the students in terms of their plausibility that θ is less than 0.6. For which student is it most plausible that Professor Ross has lived in CA for less than 60% of his lifetime? Least plausible?

2. Rank the students in terms of their answers to the question: it is equally plausible that θ is greater than or less than [blank].

3. Rank the students in terms of their uncertainty about θ. Who is the most uncertain? The least?

Example 1.4. Professor Ross will now reveal some information. After the information is revealed, reconsider your answers the questions in Example 1.2 (copied below) and revise your sketch.

1. Is it more plausible to you that θ is greater than 0.6 or less than 0.6? (In other words, it is more plausible to you that Professor Ross has lived in CA for more than 60% of his lifetime, or for less than 60% of his lifetime?) How much more plausible?

2. Repeat the previous part for a few values other than 0.6.

3. Fill in the blank: it is equally plausible that θ is greater than or less than [blank].

4. Fill in the blank: it is 2 times more plausible that θ is greater than [blank] than less than [blank].

5. Repeat the previous part for a few other ratios (e.g, 3 times).

6. What range of values accounts for most, say 98%, of your plausibility?

7. After your have considered the above questions, sketch a curve that reflects your assesment of plausibility. The plot doesn't have to perfect, but try to reflect your answers to the previous questions as best as you can.

2 Introductory Example

Statistics is the science of learning from data. But what is "Bayesian" statistics? This handout provides a relatively simple and brief example of a Bayesian statistical analysis. As you work through the example, think about: What aspects are familiar? What features are new or different? Think big picture for now; we'll fill in lots of details later.

Suppose we're interested in **the proportion of all current Cal Poly students who have ever read at least one book in the *Harry Potter* series**. We'll refer to this unknown proportion as the "population proportion", and denote it by θ (the Greek letter "theta").

We will collect data on a sample of students and use the sample data to make conclusions about θ. In particular, given the sample data, what values are more or less plausible for θ?

We'll make a plausibility assessment before collecting sample data (the "prior distribution"), and then revise our assessment after we have observed some data (the "posterior distribution").

Example 2.1. To illustrate ideas, we'll start by considering only the values 0.1, 0.3, 0.5, 0.7, 0.9 as possible for the population proportion θ. (Of course this is unrealistic; we'll consider a more practical situation later.) That is, we want to assess our relative plausibility of the statements:

- 10% of current students have read a Harry Potter book
- 30% of current students have read a Harry Potter book
- 50% of current students have read a Harry Potter book
- 70% of current students have read a Harry Potter book
- 90% of current students have read a Harry Potter book

1. Before collecting data, which of these values — 0.1, 0.3, 0.5, 0.7, 0.9 — do you think is the *least* plausible for θ?

2. Which value — 0.1, 0.3, 0.5, 0.7, 0.9 — do you think is the next least plausible? Roughly, how many *times* more plausible do you think it is than the one from the previous part?

   For example, suppose you think that 0.1 is the least plausible and 0.3 is the next least plausible value of θ. If you think that 0.3 is 1.5 times more plausible than 0.1, then, roughly, you would be willing to engage in a bet that θ is 0.1 versus 0.3 at 1.5-to-1 (or 3-to-2) odds.

3. Continue in the manner of the previous part to assign a relative plausibility to each of the values 0.1, 0.3, 0.5, 0.7, 0.9. Suppose that your least plausible value represents 1 "unit of plausibility". How many "units of plausibility" does each of the other values represent? (Assume that each of the five values has some plausibility, no matter how small; don't assign 0 plausibility to any of the values.)

4. If 0.1, 0.3, 0.5, 0.7, 0.9 are the only possible values of θ, then they should account for 100% of your total plausibility. Using your relative plausibilities from the previous part, what percent of your total plausibility does each of the values 0.1, 0.3, 0.5, 0.7, 0.9 represent? Make a table and a plot of your assessment.

Example 2.2. Now suppose we will select a random sample of $n = 15$ current students. Let y denote the number of students in the sample who have read a Harry Potter book. Before collecting data, we'll use simulation to consider what *might* happen when we select the sample.

1. If $\theta = 0.1$ what do you expect y to be? That is, if in reality 10% of *all* current Cal Poly students have read a Harry Potter book, how many students in a random sample of 15 would you expect to have read a Harry Potter book? Is this necessarily what we would see?

2. If $\theta = 0.1$, how could we use a physical object (e.g., ten-sided die, spinner) to simulate a single potential value of y?

3. If $\theta = 0.3$ what do you expect y to be? That is, if in reality 30% of *all* current Cal Poly students have read a Harry Potter book, how many students in a random sample of 15 would you expect to have read a Harry Potter book? Is this necessarily what we would see?

4. If $\theta = 0.3$, how could we use a physical object (e.g., ten-sided die, spinner) to simulate a single potential value of y?

5. We could continue in the previous way to describe how to simulate a value of y given each of the possible values of θ. Now suppose we want to simulate a single potential value of y. How should we determine which value of θ to use?

6. Describe a two-stage method for simulating (θ, y) pairs that reflects both (1) our prior plausibility assessment of θ, and (2) the natural sample-to-sample variability of values of y for given θ.

7. Conduct a few repetitions of the simulation. We'll collect our results in a plot on the board; you can sketch that plot here.

Example 2.3. So far we have considered what might happen when we collect sample data. But now we'll actually collect some data. Suppose that we observe $y = 7$ students who have read a Harry Potter book in a sample of $n = 15$ students.

Remember that we started with guesses about which values of the population proportion were more plausible than others, and we used these guesses to get a picture of what might happen in samples. Now we'll reconsider our plausibility assessment in light of the sample data that we actually observed.

1. How do you think our plausibility assessment will change given the sample data ($y = 7$, $n = 15$)?

2. How might we use the results of the simulation in the previous example to reassess our plausibility of 0.1, 0.3, 0.5, 0.7, 0.9 as values of θ given the sample data ($y = 7$, $n = 15$)?

3. Use the simulation results to sketch a plot that represents our plausibility assessment of θ given the sample data. Rank 0.1, 0.3, 0.5, 0.7, 0.9 from most to least plausible as values of θ. (See Figure 2.4 and Figure 2.5 below for a simulation with more repetitions.)

4. Compare our plausibility assessment after observing the sample data with our prior assessment. Which values are now more plausible for θ than they were before? Less? Does this make sense? Why?

Example 2.4. Now we'll look a little bit more closely at just how our posterior plausibilities were determined. This is just a brief introduction; we'll study the principles in much more depth soon.

1. Prior to observing data, how many times more plausible is 0.5 as the value of θ than 0.3?

2. Use software or a simulation applet (like this one, using left tail and right tail) to compute the probability of observing $y = 7$ in a sample of size $n = 15$ if $\theta = 0.5$. Interpret this probability.

3. Use software or a simulation applet (like this one, using left tail and right tail) to compute the probability of observing $y = 7$ in a sample of size $n = 15$ if $\theta = 0.3$. Interpret this probability.

4. How many times more likely is it to observe $y = 7$ in a sample of size $n = 15$ if $\theta = 0.5$ than if $\theta = 0.3$?

5. Based on the simulation results, what are the plausibilities of 0.5 and 0.3 after observing data? After observing data, how many times more plausible is 0.5 as the value of θ than 0.3?

6. In what relatively simple way are the three previous ratios related? (Since simulated values are only approximations, you need to think in terms of $\approx$ rather than $=$. When we introduce the underlying theory, we'll see that this is an $=$ relationship.)

7. Suggest a general principle for finding relative posterior plausibility of any two possible values of θ.

8. Given relative posterior plausibilities for all possible values of θ, how you would find the posterior plausibilities?

9. Describe how you might compute the posterior plausibilities without using simulation. Think in terms of a spreadsheet; what are the rows? What are the necessary columns and how would you fill them in? What if any value between 0 and 1 is plausible (not

just 0.1, 0.3, 0.5, 0.7, 0.9)?

2.1 Notes

```
library(tidyverse)
library(janitor)
library(kableExtra)
library(viridis)
library(scales)

bayes_col = c("#56B4E9", "#E69F00", "#009E73")
names(bayes_col) = c("prior", "likelihood", "posterior")

bayes_lty = c("dashed", "dotted", "solid")
names(bayes_lty) = c("prior", "likelihood", "posterior")
```

2.1.1 Prior plausibility

Your initial plausibility is whatever it is and reflects your background knowledge of this situation before collecting sample data. Different people will have different assessments so there are many reasonable choices.

So that we're all on the same page, suppose that before collecting sample data, our *prior* assessment of possible values of θ, from least to most plausible, is that:

- 0.1 is least plausible
- 0.3 is 4 times more plausible than 0.1
- 0.5 is 3 times more plausible than 0.3
- 0.9 is 2 times more plausible than 0.5
- 0.7 is 1.5 times more plausible than 0.9

```
# Possible values of theta
theta = seq(0.1, 0.9, by = 0.2)

# Relative plausibilities, with least plausible as baseline of 1 unit
```

3 Ideas of Bayesian Reasoning

A recent survey claims that in 2022, 35% of U.S. adults participated in "Dry January" (abstaining from alcohol for the month). Suppose we're interested in **the proportion of all current Cal Poly students who will complete a Dry January in 2023**. We'll refer to this proportion as the "population proportion" and denote it as θ (the Greek letter "theta").

Before collecting any data, consider: What do you think is the most plausible value of θ? What range of values would account for most, say 98%, of your plausibility?

Example 3.1. We'll start by considering only the values 0.1, 0.2, 0.3, 0.4, 0.5, 0.6 as initially plausible for the population proportion θ. Suppose that before collecting sample data, our *prior* assessment is that:

- 0.2 is four times more plausible than 0.1
- 0.3 is two times more plausible than 0.2
- 0.3 and 0.4 are equally plausible
- 0.2 and 0.5 are equally plausible
- 0.1 and 0.6 are equally plausible

1. Construct a table to represent this *prior distribution* of θ and sketch a plot of it. What does this distribution represent?

2. If we simulate many values of θ according to the above distribution, on what proportion of repetitions would we expect to see a value of 0.1? If we conduct 260000 repetitions of the simulation, on how many repetitions would we expect to see a value of 0.1? Repeat for the other plausible values to make a table of what we would expect the simulation results to look like. (For now, we're ignoring simulation variability; just consider the values we would expect.)

Example 3.2. The "prior" describes our initial assessment of plausibility of possible values of the population proportion θ prior to observing data. Now suppose we observe a sample of $n = 30$ students in which $y = 9$ students complete Dry January. We want to update our assessment of the population proportion in light of the observed data.

In a previous example, we used a simulation to arrive at our "posterior" plausibility. Now we'll look a little more closely at how the simulation works. Remember that to obtain the posterior from the simulation results, we conditioned on the observed data.

1. Suppose that the actual population proportion is $\theta = 0.1$. How could we use simulation to determine the likelihood of observing $y = 9$ students who have read at least one HP book in a sample of $n = 30$ students? How could we use math? (Hint: what is the probability distribution of the number of "successes" in a sample of size 30 when $\theta = 0.1$?)

2. Recall the simulation with 260000 repetitions that we started above. Consider the 10000 repetitions in which θ is 0.1. Suppose that for each of these 10000 repetitions we simulate the number of students in a sample of $n = 30$ who complete Dry January. On what proportion of these repetitions would we expect to see a sample count of $y = 9$? On how many of these 10000 repetitions would we expect to see a sample count of $y = 9$?

3. Repeat the previous parts for each of the possible values of θ: $0.1, \ldots, 0.6$. Add two columns to the table: one column for the "likelihood" of observing a count of $y = 9$ in a sample of size $n = 30$ for each value of θ, and one column for the expected number of repetitions in the simulation which would result in the count of $y = 9$ for each value of θ.

4. Consider just the repetitions that resulted in a simulated sample count of $y = 9$. What proportion of these repetitions correspond to a θ value of 0.1? Continue for the other possible values of θ to construct the *posterior distribution*, and sketch a plot of it.

5. After observing a sample of $n = 30$ Cal Poly students with a sample proportion of $y/n = 9/30 = 0.3$ who complete Dry January, what can we say about the plausibility of possible values of the *population* proportion θ? How has our assessment of plausibility changed from prior to observing the sample data? Which values are more plausible in our posterior assessment than they were in the prior? Less plausible? Does this make sense?

Example 3.3. The posterior distribution is a synthesis between the prior distribution and the observed data. Now we'll investigate more closely how the prior and the data combine to produce the posterior distribution.

1. Prior to observing data, how many times more plausible is a value of 0.3 than 0.2 for the population proportion θ?

2. Recall that we observed a sample count of $y = 9$ in a sample of size $n = 30$. How many times more likely is a count of $y = 9$ in a sample of size $n = 30$ if the population proportion θ is 0.3 than if it is 0.2?

3. After observing data, how many times more plausible is 0.3 than 0.2 for the population proportion θ?

4. How are the values from the three previous parts related?

5. Suggest a general principle for finding the posterior distribution of θ. Think in terms of a spreadsheet; what are the rows? What are the necessary columns and how would you fill them in?

Example 3.4. In the previous examples we only considered the values 0.1, 0.2, ..., 0.6 as plausible. Now we'll consider a more realistic scenario.

1. What is the problem with considering only the values 0.1, 0.2, ..., 0.6 as plausible? How could we resolve this issue?

2. Suppose we have a prior distribution that assigns initial relative plausibility to a fine grid of possible values of the population proportion θ, e.g., 0, 0.0001, 0.0002, 0.0003, ..., 0.9999, 1. We then observe a sample count of 9 in a sample of size 30. Explain the process we would use to construct a table like the one in the previous example to find the posterior distribution of the population proportion θ.

3. What are some questions of interest regarding θ? How could you use the posterior distribution to answer them? That is, once you have the posterior distribution, what would you do with it? (See Figure 3.6 for a plot of the posterior distribution.)

3.1 Notes

3.1.1 Prior distribution

```
# possible values of theta
theta = seq(0.1, 0.6, by = 0.1)

# prior plausibility
prior_units = c(1, 4, 8, 8, 4, 1)

# rescale to sum to 1
prior = prior_units / sum(prior_units)

bayes_table = data.frame(theta, prior_units, prior)

# expected simulation
n_reps = sum(prior_units) * 10000

bayes_table = bayes_table |>
  mutate(prior_reps = prior * n_reps)

bayes_table |>
  adorn_totals("row") |>
  kbl(col.names = c("Population proportion",
                    "Prior \"Units\" ",
                    "Prior",
```

4 Interpretations of Probability and Statistics

- There are two main interpretations of probability: relative frequency and "subjective" probability.
- These two interpretations provide the philosophical foundation for two schools of statistics: frequentist and Bayesian

Example 4.1. The following situations involve probability. How are the situations above similar, and how are they different? What is one feature that all of the situations have in common? Is the interpretation of "probability" the same in all situations? The goal here is to just think about these questions, and not to compute any probabilities (or to even think about how you would).

1. The probability that you roll doubles in a turn of a board game.
2. The probability that a "randomly selected" Cal Poly student is a California resident.
3. The probability that the high temperature in San Luis Obispo tomorrow is above 70 degrees F.
4. The probability that President Biden wins the 2024 U.S. Presidential Election.
5. The probability that extraterrestrial life currently exists somewhere in the universe.
6. The probability that you ate an apple on April 17, 2009.

- A phenomenon is **random** if there are multiple potential outcomes, and there is **uncertainty** about which outcome will occur. Uncertainty is understood in broad terms, and in particular does not only concern future occurrences.
- The **probability** of an event associated with a random phenomenon is a number in the interval $[0, 1]$ measuring the event's likelihood or degree of uncertainty.
- The probability of an event associated with a random phenomenon can be interpreted as a **long run proportion** or **long run relative frequency**: the probability of the event is the proportion of times that the event would occur in a very large number of hypothetical repetitions of the random phenomenon, e.g., in a simulation.

- A **subjective (a.k.a. personal) probability** describes the degree of likelihood a given individual assigns to a certain event. As the name suggests, different individuals (or probabilistic models) might have different subjective probabilities for the same event.
- **Think of subjective probabilities as measuring *relative degrees of likelihood or uncertainty*** rather than long run relative frequencies.
- We will use these interpretations interchangeably.
  - With subjective probabilities it is often helpful to consider what might happen in a simulation.
  - It is also useful to consider long run relative frequencies in terms of relative degrees of likelihood.
- Fortunately, the mathematics of probability work the same way regardless of the interpretation.

Example 4.2. What is your subjective probability that Professor Ross has a TikTok account? Consider the following two bets, and suppose you must choose only one.

A. You win $100 if Professor Ross has a TikTok account, and you win nothing otherwise.
B. A box contains 40 green and 60 gold marbles that are otherwise identical. The marbles are thoroughly mixed and one marble is selected at random. You win $100 if the selected marble is green, and you win nothing otherwise.

1. Which of the above bets would you prefer? Or are you completely indifferent? What does this say about your subjective probability that Professor Ross has a Tik Tok account?

2. If you preferred bet B to bet A, consider bet C which has a similar setup to B but now there are 20 green and 80 gold marbles. Do you prefer bet A or bet C? What does this say about your subjective probability that Professor Ross has a Tik Tok account?

3. If you preferred bet A to bet B, consider bet D which has a similar setup to B but now there are 60 green and 40 gold marbles. Do you prefer bet A or bet D? What does this say about your subjective probability that Professor Ross has a Tik Tok account?

4. Continue to consider different numbers of green and gold marbles. Can you zero in on your subjective probability?

Example 4.3. Suppose your subjective probabilities for the 2023 NBA champion satisfy the following conditions.

- The Nuggets and 76ers are equally likely to win
- The Grizzlies are 1.5 times more likely than the Nuggets to win
- The Celtics are 2 times more likely than the Grizzlies to win
- The winner is as likely to be among these four teams — Celtics, Grizzlies, 76ers, Nuggets — as not.

1. Construct a table of your subjective probabilities.

2. How could you construct a circular spinner (like from a kids game) to simulate the NBA champion according to these subjective probabilities? According to this model, what would you expect the results of 10000 repetitions of a simulation of the champion to look like?

Example 4.4. How old do you think Professor Ross currently is? Consider age on a continuous scale, e.g., you might be 20.73 or 21.36 or 19.50.

In this example, you will use probability to quantify your uncertainty about your instructor's age. You only need to give ballpark estimates of your subjective probabilities, but you might consider what kinds of bets you would be willing to accept.

1. What is your subjective probability that your instructor is at most 30 years old? More than 30 years old? (What must be true about these two probabilities?)

2. What is your subjective probability that your instructor is at most 60 years old? More than 60 years old?

3. What is your subjective probability that your instructor is at most 40 years old? More than 40 years old?

4. What is your subjective probability that your instructor is at most 50 years old? More than 50 years old?

5. Fill in the blank: your subjective probability that your instructor is at most [blank] years old is equal to 50%.

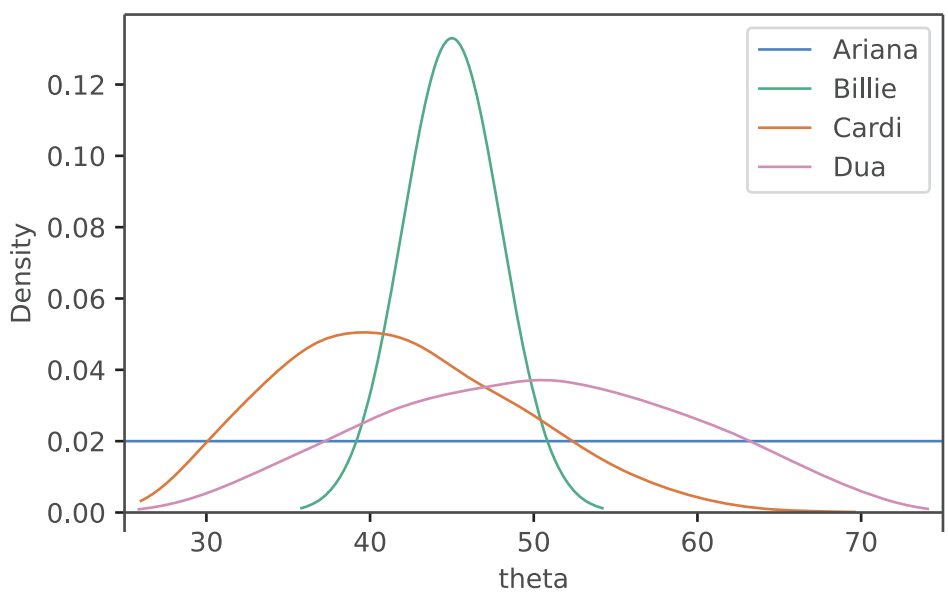
6. Fill in the blanks: your subjective probability that your instructor is between [blank] and [blank] years old is equal to 95%.

7. Let θ represent your instructor's age at 8:00am on Jan 12, 2023. Use your answers to the previous parts to sketch a continuous probability density function to represent your *subjective probability distribution* for θ.

8. If you ascribe a probability distribution to θ, then are you treating θ as a constant or a random variable?

- A **random variable** is a numerical quantity whose value is determined by the outcome of a random or uncertain phenomenon.
- The **(probability) distribution** of a random variable specifies the possible values of the random variable and a way of determining corresponding probabilities. Like probabilities themselves, probability distributions of random variables can also be interpreted as:
  - *relative frequency distributions*, e.g., what pattern would emerge if I simulated many values of the random variable? or as
  - *subjective probability distributions*, e.g., which potential values of this uncertain quantity are relatively more plausible than others?

Example 4.5. Continuing the previous example, the figure displays the subjective probability distribution of the instructor's age for four students.



1. Since age is treated as a continuous random variable, each of the above plots is a probability "density". Explain briefly what this means. How is probability represented

in density plots like these?

2. Rank the students in terms of their subjective probability that the instructor is at most 40 years old.

3. Rank the students in terms of their answers to the question: your subjective probability that your instructor is at most [blank] years old is equal to 50%.

4. Rank the students in terms of their uncertainty about the instructor's age. Who is the most uncertain? The least?

- One key aspect of Bayesian analyses is *applying a subjective probability distribution to a parameter* in a statistical model.

Example 4.6. Let θ_b represent the proportion of current Cal Poly students who have ever read any of the books in the *Harry Potter* series. Let θ_m represent the proportion of current Cal Poly students who have ever seen any of the movies in the *Harry Potter* series.

1. Are θ_b and θ_m parameters or statistics? Why?

2. Are the values of θ_b and θ_m known or unknown, certain or uncertain?

3. What are the possible values of θ_b and θ_m?

4. Sketch a probability distribution representing what you think are more/less credible values of θ_b. Repeat for θ_m.

5. Are you more certain about the value of θ_b or θ_m? How is this reflected in your distributions?

6. Suppose that in a class of 35 Cal Poly students, 21 have read a Harry Potter book, and 30 have seen a Harry Potter movie. Now that we have observed some data, sketch a probability distribution representing what you think are more/less credible values of θ_b. Repeat for θ_m. How do your distributions after observing data compare to the distributions you sketched before?

- **Observational units** (a.k.a., cases, individuals, subjects) are the people, places, things, etc we collect information on.
- A **variable** is any characteristic of an observational unit that we can measure.
- **Statistical inference** involves using data collected on a *sample* to make conclusions about a *population*.
- Inference often concerns specific numerical summaries, using values of *statistics* to make conclusions about *parameters*.
- A **statistic** is a number that describes the **sample**, e.g., *sample mean*, *sample proportion*.

- A **parameter** is a number that describes the **population** or model, e.g., *population mean*, *population proportion*. The actual value of a parameter is almost always *unknown*.
  - Parameters are often denoted with Greek letters. We'll often use the Greek letter θ ("theta") to denote a generic parameter.
- In "traditional", *frequentist* statistical analysis, parameters are treated as *fixed — that is, not random — constants*.
  - Any randomness in a frequentist analysis arises from how the data were collected, e.g., via random sampling or random assignment.
  - In a frequentist analysis, statistics are random variables; parameters are fixed numbers.
- On the other hand, in a Bayesian statistical analysis, since a parameter θ is unknown — that is, it's value is *uncertain* to the observer — θ is treated as a *random variable*.
  - That is, **in Bayesian statistical analyses unknown parameters are random variables that have probability distributions.**
  - The (subjective) probability distribution of a parameter quantifies the degree of uncertainty about the value of the parameter.
  - Therefore, the Bayesian perspective allows for probability statements about parameters.
  - We start with "prior" distributions that represent our uncertainty based on our "beliefs", then we revise to obtain "posterior" distributions after observing some data.
  - We will see (among other things) (1) how to quantify uncertainty about parameters using probability distributions, and (2) how to update those distributions to reflect new data.

5 Bayes Rule

The mechanism that underpins all of Bayesian statistical analysis is *Bayes' rule*, which describes how to update uncertainty in light of new information, evidence, or data.

Example 5.1. A recent survey of American adults asked: "Based on what you have heard or read, which of the following two statements best describes the scientific method?"

- 70% selected "The scientific method produces findings meant to be continually tested and updated over time". (We'll call this the "iterative" opinion.)
- 14% selected "The scientific method identifies unchanging core principles and truths". (We'll call this the "unchanging" opinion).
- 16% were not sure which of the two statements was best.

How does the response to this question change based on education level? Suppose education level is classified as: high school or less (HS), some college but no Bachelor's degree (college), Bachelor's degree (Bachelor's), or postgraduate degree (postgraduate). The education breakdown is

- Among those who agree with "iterative": 31.3% HS, 27.6% college, 22.9% Bachelor's, and 18.2% postgraduate.
- Among those who agree with "unchanging": 38.6% HS, 31.4% college, 19.7% Bachelor's, and 10.3% postgraduate.
- Among those "not sure": 57.3% HS, 27.2% college, 9.7% Bachelor's, and 5.8% postgraduate

1. Use the information to construct an appropriate two-way table.

2. Overall what percentage of adults have a postgraduate degree? How is this related to the values 18.2%, 10.3%, and 5.8%?

3. What percent of those with a postgraduate degree agree that the scientific method is "iterative"? How is this related to the values provided?

Bayes' rule for events specifies how a prior probability $P(H)$ of event H is updated in response to the evidence E to obtain the posterior probability $P(H|E)$.

$$P(H|E) = \frac{P(E|H)P(H)}{P(E)}$$

- Event H represents a particular hypothesis (or model or case)
- Event E represents observed evidence (or data or information)
- $P(H)$ is the unconditional or **prior probability** of H (prior to observing E)
- $P(H|E)$ is the conditional or **posterior probability** of H after observing evidence E.
- $P(E|H)$ is the **likelihood** of evidence E given hypothesis (or model or case) H

Example 5.2. Continuing Example 5.1. Randomly select an American adult.

1. Consider the conditional probability that a randomly selected American adult agrees that the scientific method is "iterative" given that they have a postgraduate degree. Identify the prior probability, hypothesis, evidence, likelihood, and posterior probability, and use Bayes' rule to compute the posterior probability.

2. Find the conditional probability that a randomly selected American adult with a postgraduate degree agrees that the scientific method is "unchanging".

3. Find the conditional probability that a randomly selected American adult with a postgraduate degree is not sure about which statement is best.

4. How many times more likely is it for an *American adult* to have a postgraduate degree and agree with the "iterative" statement than to have a postgraduate degree and agree with the "unchanging" statement?

5. How many times more likely is it for an *American adult with a postgraduate degree* to agree with the "iterative" statement than to agree with the "unchanging" statement?

6. What do you notice about the answers to the two previous parts?

7. How many times more likely is it for an *American adult* to agree with the "iterative" statement than to agree with the "unchanging" statement?

8. How many times more likely is it for an American adult to have a postgraduate degree when the adult agrees with the "iterative" statement than when the adult agree with the "unchanging" statement?

9. How many times more likely is it for an *American adult with a postgraduate degree* to agree with the "iterative" statement than to agree with the "unchanging" statement?

10. How are the values in the three previous parts related?

Bayes rule is often used when there are multiple hypotheses or cases. Suppose $H_1, \ldots, H_k$ is a series of distinct hypotheses which together account for all possibilities, and E is any event (evidence). Then Bayes' rule implies that the posterior probability of any particular hypothesis H_j satisfies

$$P(H_j|E) = \frac{P(E|H_j)P(H_j)}{P(E)}$$

The marginal probability of the evidence, $P(E)$, in the denominator can be calculated using the *law of total probability*

$$P(E) = \sum_{i=1}^{k} P(E|H_i)P(H_i)$$

The law of total probability says that we can interpret the unconditional probability $P(E)$ as a probability-weighted average of the case-by-case conditional probabilities $P(E|H_i)$ where the weights $P(H_i)$ represent the probability of encountering each case.

Combining Bayes' rule with the law of total probability,

$$\begin{aligned} P(H_j|E) &= \frac{P(E|H_j)P(H_j)}{P(E)} \\ &= \frac{P(E|H_j)P(H_j)}{\sum_{i=1}^{k} P(E|H_i)P(H_i)} \\ P(H_j|E) &\propto P(E|H_j)P(H_j) \end{aligned}$$

In short, Bayes' rule says

$$\textbf{posterior} \propto \textbf{likelihood} \times \textbf{prior}$$

Bayes rule calculations are often organized in a **Bayes' table** which illustrates "posterior is proportional to likelihood times prior". The table has one row for each hypothesis and columns for

- prior probability: column sum is 1
- likelihood of the evidence given each hypothesis
  - likelihood depends on the evidence; if the evidence changes, the likelihood column changes
  - the sum of the likelihood column is a meaningless number and can be any value
- product of prior and likelihood: column sum is the marginal probability of the evidence
- posterior probability: column sum is 1

The process of conditioning can be thought of as **"slicing and renormalizing".**

- Extract the "slice" corresponding to the event being conditioned on (and discard the rest). For example, a slice might correspond to a particular row or column of a two-way table.

- "Renormalize" the values in the slice so that corresponding probabilities add up to 1.

Example 5.3. Continuing Example 5.1. Randomly select an American adult. Now suppose we want to compute the posterior probabilities for an American adult's perception of the scientific method given that the randomly selected American adult has some college but no Bachelor's degree ("College").

1. Before computing, make an educated guess for the posterior probabilities. In particular, will the changes from prior to posterior be more or less extreme given the American has some college but no Bachelor's degree than when given the American has a postgraduate degree? Why?

2. Construct a Bayes table and compute the posterior probabilities. Compare to the posterior probabilities given postgraduate degree from the previous examples.

Example 5.4. Suppose that you are presented with six boxes, labeled 0, 1, 2, ..., 5, each containing five marbles. Box 0 contains 0 green and 5 gold marbles, box 1 contains 1 green and 4 gold, and so on with box i containing i green and $5-i$ gold. One of the boxes is chosen uniformly at random (perhaps by rolling a fair six-sided die), and then you will randomly select marbles from that box, without replacement. Based on the colors of the marbles selected, you will update the probabilities of which box had been chosen.

1. Suppose that a single marble is selected and it is green. Which box do you think is the most likely to have been chosen? Make a guess for the posterior probabilities for each box. Then construct a Bayes table to compute the posterior probabilities. How do they compare to the prior probabilities?

2. Now suppose a second marble is selected from the same box, without replacement, and its color is gold. Which box do you think is the most likely to have been chosen given these two marbles? Make a guess for the posterior probabilities for each box. Then construct a Bayes table to compute the posterior probabilities, *using the posterior probabilities from the previous part after the selection of the green marble as the new prior probabilities before seeing the gold marble.*

3. Now construct a Bayes table corresponding to the original prior probabilities (1/6 each) and the combined evidence that the first ball selected was green and the second was gold. How do the posterior probabilities compare to the previous part?

4. In the previous part, the first ball selected was green and the second was gold. Suppose you only knew that in a sample of two marbles, 1 was green and 1 was gold. That

is, you didn't know which was first or second. How would the previous part change? Should knowing the order matter? Does it?

- Bayesian analyses are often performed sequentially.
- Posterior probabilities can be sequentially updated as new data becomes available, with the posterior probabilities after the previous stage serving as the prior probabilities for the next stage.
- The final posterior probabilities only depend upon the cumulative data. It doesn't matter if we sequentially update the posterior after each new piece of data or only once after all the data is available; the final posterior probabilities will be the same either way.
- Also, the final posterior probabilities are not impacted by the order in which the data are observed.

5.1 Notes

5.1.1 Two-way table

```
hypothesis = c("iterative", "unchanging", "not sure")

prior = c(0.70, 0.14, 0.16)

evidence = c("HS", "College", "Bachelors", "Postgrad")

likelihood_E_given_H1 = c(0.313, 0.276, 0.229, 0.182) # given iterative
likelihood_E_given_H2 = c(0.386, 0.314, 0.197, 0.103) # given unchanging
likelihood_E_given_H3 = c(0.573, 0.272, 0.097, 0.058) # given not sure

n = 100000

twoway_table = data.frame(n * rbind(
             prior[1] * likelihood_E_given_H1,
             prior[2] * likelihood_E_given_H2,
             prior[3] * likelihood_E_given_H3))

colnames(twoway_table) = evidence
```

6 Introduction to Estimation

- A *parameter* is a number that describes the population, e.g., population mean, population proportion. The actual value of a parameter is almost always unknown.
- A *statistic* is a number that describes the sample, e.g., sample mean, sample proportion.
- Observed sample statistics can be used to *estimate* unknown population parameters.
- **Bayesian estimation**
  - Regards parameters as *random variables* with probability distributions
  - Assigns a subjective **prior distribution** to parameters
  - *Conditions* on the observed data
  - Determines parameter estimates based on the **posterior distribution**, which is obtained via Bayes rule

$$\text{posterior} \propto \text{prior} \times \text{likelihood}$$

- For given data y, the **likelihood function** $f(y|\theta)$ is the probability (or density for continuous data) of observing the sample data y viewed as a *function of the parameter* θ.
  - In the likelihood function, the observed value of the data y is treated as a fixed constant; the likelihood is a function of parameters θ.
  - The likelihood function is used in both Bayesian and frequentist statistics, but in different ways[1].

Example 6.1. Most people are right-handed, and even the right eye is dominant for most people. In a 2003 study reported in *Nature*, a German bio-psychologist conjectured that this preference for the right side manifests itself in other ways as well. In particular, he investigated if people have a tendency to lean their heads to the right when kissing. The researcher observed kissing couples in public places and recorded whether the couple leaned their heads to the right or left. (We'll assume this represents a randomly selected representative sample of kissing couples.)

[1] For example, maximum likelihood estimation is a common *frequentist* technique for estimating the value of a parameter based on data from a sample. The value of a parameter that maximizes the likelihood function is called a *maximum likelihood estimate* (MLE). The MLE depends on the data y: the MLE is the value of θ which gives the largest likelihood of having produced the observed data y.

The parameter of interest is the *population proportion of kissing couples who lean their heads to the right*. Denote this unknown parameter θ; our goal is to estimate θ based on sample data. We'll take a Bayesian approach to estimating θ: we'll treat the unknown parameter θ as a *random variable* and wish to find its posterior distribution after observing data on a sample of n kissing couples of which y lean their heads to the right.

1. Sketch a distribution representing your plausibility assessment of θ prior to observing the sample data.

2. We will start with a very simplified, unrealistic prior distribution that assumes only five possible, equally likely values for θ: 0.1, 0.3, 0.5, 0.7, 0.9. Sketch the prior distribution and start constructing a Bayes table.

3. Now suppose that we observe a sample of $n = 12$ kissing couples of which $y = 8$ lean right. Describe in principle would you fill in the likelihood column.

4. If $\theta = 0.1$ what is the distribution of y? Compute and interpret the probability that $y = 8$ if $\theta = 0.1$.

5. If $\theta = 0.3$ what is the distribution of y? Compute and interpret the probability that $y = 8$ if $\theta = 0.3$.

6. If $\theta = 0.5$ what is the distribution of y? Compute and interpret the probability that $y = 8$ if $\theta = 0.5$.

7. If $\theta = 0.7$ what is the distribution of y? Compute and interpret the probability that $y = 8$ if $\theta = 0.7$.

8. If $\theta = 0.9$ what is the distribution of y? Compute and interpret the probability that $y = 8$ if $\theta = 0.9$.

9. Sketch a plot of the likelihood function and fill in the likelihood column in the Bayes table.

10. Complete the Bayes table and sketch a plot of the posterior distribution.

11. Examine the posterior distribution. What does it say about θ? How does it compare to the prior and the likelihood?

12. Now consider a prior distribution which places probability 1/9, 2/9, 3/9, 2/9, 1/9 on the values 0.1, 0.3, 0.5, 0.7, 0.9, respectively. Assuming the same sample data ($y/n = 8/12$), construct a Bayes table using this prior. How does the posterior distribution compare to the one based on the equally likely prior?

13. Now consider a prior distribution which places probability 5/15, 4/15, 3/15, 2/15, 1/15 on the values 0.1, 0.3, 0.5, 0.7, 0.9, respectively. Assuming the same sample data ($y/n = 8/12$), construct a Bayes table using this prior. How does the posterior distribution compare to the ones from the previous parts?

- In Bayesian estimation parameters are treated as *random variables* with probability distributions[2].
- The *posterior distribution* contains all relevant information about parameters after observing sample data. That is, all Bayesian inference is based on the posterior distribution.
- The posterior distribution is a compromise between
  - prior "beliefs/plausibility", as represented by the prior distribution
  - data, as represented by the likelihood function

Example 6.2. Continuing Example 6.1. While the previous exercise introduced the main ideas, it was unrealistic to consider only five possible values of θ.

1. What are the *possible* values of θ? Does the *parameter* θ take values on a continuous or discrete scale? (Careful: we're talking about the parameter and not the data.)

[2]In contrast, a frequentist approach regards parameters as unknown but fixed (not random) quantities.

2. Let's assume that any multiple of 0.0001 is a possible value of θ: $0, 0.0001, 0.0002, \ldots, 0.9999, 1$. Assume a discrete uniform (equally likely) prior distribution on these values. Suppose again that $y = 8$ couples in a sample of $n = 12$ kissing couples lean right. Use software to plot the prior distribution, the (scaled) likelihood function, and then find the posterior and plot it. Describe the posterior distribution. What does it say about θ?

3. Now assume a prior distribution which is proportional to $1 - 2|\theta - 0.5|$ for $\theta = 0, 0.0001, 0.0002, \ldots, 0.9999, 1$. Use software to plot this prior; what does it say about θ? Then suppose again that $y = 8$ couples in a sample of $n = 12$ kissing couples lean right. Use software to plot the prior distribution, the (scaled) likelihood function, and then find the posterior and plot it. What does the posterior distribution say about θ?

4. Now assume a prior distribution which is proportional to $1-\theta$ for $\theta = 0, 0.0001, 0.0002, \ldots, 0.9999, 1$. Use software to plot this prior; what does it say about θ? Then suppose again that $y = 8$ couples in a sample of $n = 12$ kissing couples lean right. Use software to plot the prior distribution, the (scaled) likelihood function, and then find the posterior and plot it. What does the posterior distribution say about θ?

5. Compare the posterior distributions corresponding to the three different priors. How does each posterior distribution compare to the prior and the likelihood? Does the prior distribution influence the posterior distribution?

- Even in situations where the data are discrete (e.g., binary success/failure data, count data), most statistical *parameters* take values on a *continuous* scale.
- Thus in a Bayesian analysis, parameters are usually *continuous random variables*, and have *continuous probability distributions*, a.k.a., *densities*.
- An alternative to dealing with continuous distributions is to use **grid approximation**: Treat the parameter as discrete, on a sufficiently fine grid of values, and use discrete distributions.

Example 6.3. Continuing Example 6.2. Now we'll perform a Bayesian analysis on the actual study data in which $y = 80$ couples out of a sample of $n = 124$ leaned right. We'll again use a grid approximation and assume that any multiple of 0.0001 between 0 and 1 is a possible value of θ: $0, 0.0001, 0.0002, \ldots, 0.9999, 1$.

1. Before performing the Bayesian analysis, use software to plot the likelihood when $y = 80$ couples in a sample of $n = 124$ kissing couples lean right. How does the likelihood for this sample compare to the likelihood based on the smaller sample (8/12) from previous exercises?

2. Now back to Bayesian analysis. Assume a discrete uniform prior distribution for θ. Suppose that $y = 80$ couples in a sample of $n = 124$ kissing couples lean right. Use software to plot the prior distribution, the likelihood function, and then find the posterior and plot it. Describe the posterior distribution. What does it say about θ?

3. Now assume a prior distribution which is proportional to $1 - 2|\theta - 0.5|$ for $\theta = 0, 0.0001, 0.0002, \ldots, 0.9999, 1$. Then suppose again that $y = 80$ couples in a sample of $n = 124$ kissing couples lean right. Use software to plot the prior distribution, the likelihood function, and then find the posterior and plot it. What does the posterior distribution say about θ?

4. Now assume a prior distribution which is proportional to $1-\theta$ for $\theta = 0, 0.0001, 0.0002, \ldots, 0.9999, 1$. Then suppose again that $y = 80$ couples in a sample of $n = 124$ kissing couples lean right. Use software to plot the prior distribution, the likelihood function, and then find the posterior and plot it. What does the posterior distribution say about θ?

5. Compare the posterior distributions corresponding to the three different priors. How does each posterior distribution compare to the prior and the likelihood? Comment on the influence that the prior distribution has. Does the Bayesian inference for this data appear to be highly sensitive to the choice of prior? How does this compare to the $n = 12$ situation?

6. If you had to produce a single number Bayesian estimate of θ based on the sample data, what number might you pick?

- In a Bayesian analysis, the posterior distribution contains all relevant information about parameters after observing sample data.
- We often use certain summary characteristics of the posterior distribution to make inferences about parameters.
- A **point estimate** of an unknown parameter is a single-number estimate of the parameter.
- Given a posterior distribution of a parameter θ, three commonly used Bayesian point estimates of θ are:
  - the posterior mean
  - the posterior median
  - the posterior mode.

- In many situations, the posterior distribution will be roughly symmetric with a single peak, in which case posterior mean, median, and mode will all be about the same.
- Reducing the posterior distribution to a single-number point estimate loses a lot of the information the posterior distribution provides. The entire posterior distribution quantifies the uncertainty about θ after observing sample data. We will soon see how to more fully use the posterior distribution in making inference about θ.

Example 6.4. Continuing Example 6.1 where θ can only take values 0.1, 0.3, 0.5, 0.7, 0.9. Consider a prior distribution which places probability 5/15, 4/15, 3/15, 2/15, 1/15 on the values 0.1, 0.3, 0.5, 0.7, 0.9, respectively. Ssuppose that $y = 8$ couples in a sample of size $n = 12$ lean right. Recall the Bayes table.

1. Suppose we want a single number point estimate of θ *before* observing sample data. Find the mode of the prior distribution of θ, a.k.a., the "prior mode".

2. Find the expected value of the prior distribution of θ, a.k.a., the "prior mean".

3. Suppose we want a single number point estimate of θ *after* observing sample data. Find the mode of the posterior distribution of θ, a.k.a., the "posterior mode".

4. Find the expected value of the prior distribution of θ, a.k.a., the "posterior mean". How does the posterior mean compare to the prior mean and the observed sample propotion?

7 Introduction to Inference

- In a Bayesian analysis, the posterior distribution contains all relevant information about parameters after observing sample data.
- We can use the posterior distribution to make inferences about parameters.

Example 7.1. Suppose we want to estimate θ, the population proportion of American adults who have read a book in the last year.

1. Sketch your prior distribution for θ.

2. Suppose Frog formulates a Normal distribution prior for θ. Frog's prior mean is 0.4 and prior standard deviation is 0.1. What are some things that Frog's prior says about θ?

3. Suppose Toad formulates a Normal distribution prior for θ. Toad's prior mean is 0.4 and prior standard deviation is 0.05. Who has more prior certainty about θ? Why?

- The **prior standard deviation** summarizes in a single number the degree of uncertainty about θ before observing sample data.
- The **posterior standard deviation** summarizes in a single number the degree of uncertainty about θ after observing sample data.

- The smaller the posterior (prior) standard deviation, the more certainty we have about the value of the parameter after (before) observing sample data.

Example 7.2. Continuing Example 7.1, we'll assume a Normal prior distribution for θ with prior mean 0.4 and prior standard deviation 0.1.

1. Compute and interpret the prior probability that θ is greater than 0.7.

2. Find the 25th and 75th percentiles of the prior distribution. What is the prior probability that θ lies in the interval with these percentiles as endpoints? According to the prior, how plausible is it for θ to lie inside this interval relative to outside it?

3. Repeat the previous part with the 10th and 90th percentiles of the prior distribution.

4. Repeat the previous part with the 1st and 99th percentiles of the prior distribution.

Example 7.3. Continuing Example 7.1, we'll assume a Normal prior distribution for θ with prior mean 0.4 and prior standard deviation 0.1.

In a sample of 150 American adults, 75% have read a book in the past year. (The 75% value is motivated by a real sample we'll see in a later example.)

1. Use grid approximation to approximate the posterior distribution. Make a plot of prior, likelihood, and posterior. Describe the shape of the posterior distribution, and approximate the posterior mean and posterior SD.

2. The posterior distribution is approximately Normal with posterior mean 0.709 and posterior SD 0.036. How does the sample mean compare, roughly, to the prior mean and the observed sample proportion?

3. How does the posterior standard deviation compare to the prior standard deviation? Why?

4. Compute and interpret the posterior probability that θ is greater than 0.7. Compare to the prior probability.

5. Find the 25th and 75th percentiles of the posterior distribution. What is the posterior probability that θ lies in the interval with these percentiles as endpoints? According to the posterior, how plausible is it for θ to lie inside this interval relative to outside it? Compare to the prior interval.

6. Repeat the previous part with the 10th and 90th percentiles of the posterior distribution.

7. Repeat the previous part with the 1st and 99th percentiles of the posterior distribution.

- Bayesian inference for a parameter is based on its posterior distribution.
- Since a Bayesian analysis treats parameters as random variables, *it is possible to make probability statements about a parameter*.
- A Bayesian **credible interval** is an interval of values for the parameter that has at least the specified probability, e.g., 50%, 80%, 98%.
  - With a 50% credible interval, it is equally plausible that the parameter lies inside the interval as outside
  - With an 80% credible interval, it is 4 times more plausible that the parameter lies inside the interval than outside
  - With a 98% credible interval, it is 49 times more plausible that the parameter lies inside the interval than outside
- **Central credible intervals** split the complementary probability evenly between the two tails.
  - The endpoints of a 50% central posterior credible interval are the 25th and the 75th percentiles of the posterior distribution.
  - The endpoints of an 80% central posterior credible interval are the 10th and the 90th percentiles of the posterior distribution.
  - The endpoints of a 98% central posterior credible interval are the 1st and the 99th percentiles of the posterior distribution.
- Bayesian inference is based on the full posterior distribution. Credible intervals simply provide a summary of this distribution.
- Reporting a few credible intervals, rather than just one, provides a richer picture of how the posterior distribution represents the uncertainty in the parameter.
- In many situations, the posterior distribution of a single parameter is approximately Normal, so posterior probabilities can be approximated with Normal distribution calculations. If the posterior distribution of a parameter is approximately Normal, an

approximate central credible interval has endpoints

$$\text{posterior mean} \pm z^* \times \text{posterior SD}$$

where the multiple (a.k.a., critical value) which corresponds to the credibility level comes from a Normal distribution, e.g.,

Central credibility	50%	80%	95%	98%
Normal z^* multiple	0.67	1.28	1.96	2.33

Example 7.4. Continuing Example 7.1, we'll assume a Normal prior distribution for θ with prior mean 0.4 and prior standard deviation 0.1.

In a recent survey of 1502 American adults conducted by the Pew Research Center, 75% of those surveyed said they have read a book in the past year.

1. Use grid approximation to approximate the posterior distribution. Make a plot of prior, likelihood, and posterior. Describe the shape of the posterior distribution, and approximate the posterior mean and posterior SD. How does this posterior compare to the one based on the smaller sample size ($n = 150$)?

2. The posterior distribution is approximately Normal with posterior mean 0.745 and posterior SD 0.011. Compute *and interpret in context* the posterior probability that θ is greater than 0.7. Compare to the prior probability, and to the posterior probability based on the smaller sample size ($n = 150$).

3. Compute and *interpret in context* 50%, 80%, and 98% central posterior credible intervals for θ. Compare to the prior intervals, and to the posterior intervals based on the smaller sample size ($n = 150$).

4. Here is how the survey question was worded: "During the past 12 months, about how many BOOKS did you read either all or part of the way through? Please include any print, electronic, or audiobooks you may have read or listened to." Does this change your conclusions? Explain.

- The quality of any statistical analysis depends very heavily on the quality of the data.
- Always investigate how the data were collected to determine what conclusions are appropriate.
- Is the sample reasonably representative of the population? Were the variables reliably measured?

Example 7.5. Continuing Example 7.4, we'll use the same sample data ($n = 1502$, 75%) but now we'll consider different priors.

For each of the priors below, plot prior, likelihood, and posterior, and compute the *posterior* probability that θ is greater than 0.7. Compare.

1. Normal distribution prior with prior mean 0.4 and prior SD 0.05.

2. Uniform distribution prior on the interval [0, 0.7]

- You have a great deal of flexibility in choosing a prior, and there are many reasonable approaches.
- However, do NOT choose a prior that assigns 0 probability/density to possible values of the parameter regardless of how initially implausible the values are.

8 Introduction to Inference for a Population Mean

- Recall that the likelihood function is the probability (or density for continuous data) of observing the sample data y viewed as a *function of the parameter* θ.
- When the data y takes values on a continuous scale, the likelihood is determined by the *probability density function* of y given θ, $f(y|\theta)$.
- In the likelihood function, the observed value of the data y is treated as a fixed constant, and the likelihood of observing that y is evaluated for all potential values of θ.

Example 8.1. Assume body temperatures (degrees Fahrenheit) of healthy adults follow a Normal distribution with unknown mean θ and known standard deviation $\sigma = 1$. (It's unrealistic to assume the population standard deviation is known. We'll consider the case of unknown standard deviation later.) Suppose we wish to estimate θ, the population mean healthy human body temperature.

1. If θ was known (say $\theta = 98.6$), what would the distribution of y look like? What are some of its features? What does the standard deviation $\sigma = 1$ represent?

2. Sketch your prior distribution for θ. What is your single most plausible value for θ? About what range of values accounts for most (say 98%) of your prior plausibility? Roughly, what is your prior standard deviation? What does your prior standard deviation represent?

Be careful to distinguish between

- Model for the data
  - The conditional distribution of the measured variables y given the parameters θ, which determines (once the data are observed) the likelihood
  - The model for the data represents our assumptions about how the measured variables y vary from unit-to-unit in the population
- Prior model for parameters
  - The prior (marginal) distribution of the parameters θ which are present in the model for the data
  - The prior distribution represents — via a probability distribuion — our assessment of plausibility/uncertainty of values of parameters θ prior to observing sample data

Example 8.2. Continuing Example 8.1. Assume first a discrete prior distribution for θ which places probability 0.10, 0.25, 0.30, 0.25, 0.10 on the values 97.6, 98.1, 98.6, 99.1, 99.6, respectively.

1. Suppose a single temperature value of 97.9 is observed. Construct a Bayes table and find the posterior distribution of θ. In particular, how do you determine the likelihood?

2. Now suppose a second temperature value, 97.5, is observed, independently of the first. Construct a Bayes table and find the posterior distribution of θ after observing these two measurements, using the posterior distribution from the previous part as the prior distribution in this part.

3. Now consider the original prior again. Determine the likelihood of observing temperatures of 97.9 and 97.5 in a sample of size 2. Then construct a Bayes table, starting with the original prior, and find the posterior distribution of θ after observing these two measurements. Compare to the previous part.

4. Consider the original prior again. Suppose that we take a random sample of two temperature measurements, but instead of observing the two individual values, we only observe that the sample mean is 97.7. Determine the likelihood of observing a sample mean of 97.7 in a sample of size 2. (Hint: if $\bar{y}$ is the sample mean of n values from a $N(\theta, \sigma)$ distribution, what is the distribution of $\bar{y}$?) Then construct a Bayes table and find the posterior distribution of θ after observing this sample mean 97.7. Compare to the previous part.

- It is often not necessary to know all the individual data values to evaluate the *shape* of the likelihood as a function of the parameter θ, but rather simply the values of a few summary statistics.

Example 8.3. Continuing Example 8.1. We'll now use a grid approximation and assume that any multiple of 0.0001 between 96.0 and 100.0 is a possible value of θ: 96.0, 96.0001, 96.0002, ..., 99.9999, 100.0.

Suppose that the sample mean temperature is $\bar{y} = 97.7$ in a sample of $n = 2$ temperature measurements.

1. Assume a discrete uniform prior distribution over θ values in the grid. Use software to plot the prior distribution, the (scaled) likelihood function, and then find the posterior and plot it. Describe the posterior distribution. What does it say about θ?

2. Assume a discrete triangular prior distribution which is proportional to $2 - |\theta - 98.0|$ over θ values in the grid. Use software to plot the prior distribution, the (scaled) likelihood function, and then find the posterior and plot it. Describe the posterior distribution. What does it say about θ?

3. Now assume a prior distribution which is proportional to a Normal distribution with mean 98.6 and standard deviation 0.7 over θ values in the grid. (Careful: be sure not to confuse a Normal prior distribution with the likelihood.) Use software to plot the prior distribution, the (scaled) likelihood function, and then find the posterior and plot it. Describe the posterior distribution. What does it say about θ?

4. Compare the posterior distributions corresponding to the different priors. How does each posterior distribution compare to the prior and the likelihood? Comment on the influence that the prior distribution has. (But remember, this is only a sample of size $n = 2$.)

Example 8.4. Continuing Example 8.1. In a recent study[1], the sample mean body temperature in a sample of 208 healthy adults was 97.7 degrees F.

We'll again use a grid approximation and assume that any multiple of 0.0001 between 96.0 and 100.0 is a possible value of θ: $96.0, 96.0001, 96.0002, \ldots, 99.9999, 100.0$.

1. Assume a discrete uniform prior distribution over θ values in the grid. Use software to plot the prior distribution, the (scaled) likelihood function, and then find the posterior and plot it. Describe the posterior distribution. What does it say about θ?

2. Assume a discrete triangular prior distribution which is proportional to $2 - |\theta - 98.0|$ over θ values in the grid. Use software to plot the prior distribution, the (scaled) likelihood function, and then find the posterior and plot it. Describe the posterior

[1] Source and a related article.

distribution. What does it say about θ?

3. Now assume a prior distribution which is proportional to a Normal distribution with mean 98.6 and standard deviation 0.7 over θ values in the grid. (Careful: be sure not to confuse a Normal prior distribution with the likelihood.) Use software to plot the prior distribution, the (scaled) likelihood function, and then find the posterior and plot it. Describe the posterior distribution. What does it say about θ?

4. Compare the posterior distributions corresponding to the three different priors. How does each posterior distribution compare to the prior and the likelihood? Comment on the influence that the prior distribution has; how does this compare to the $n = 2$ situation?

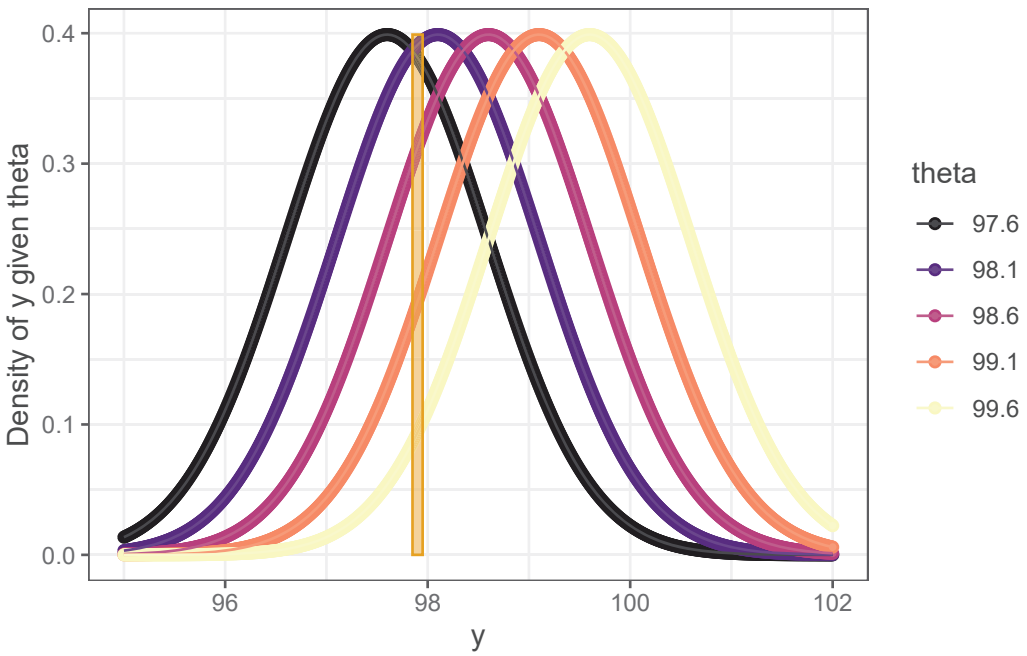
5. The posterior distribution is approximately Normal with posterior mean 97.71 and posterior SD 0.07. Explain what the posterior SD represents.

6. Compute *and interpret in context* the posterior probability that θ is less than 98.6. Compare to the prior probability.

7. Compute and *interpret in context* 50%, 80%, and 98% central posterior credible intervals for θ. Compare to the prior intervals.

8.1 Notes

8.1.1 Likelihood of single value



9 Comparing Bayesian and Frequentist Analysis

The most widely used elements of "traditional" frequentist inference are confidence intervals and hypothesis tests (a.k.a, null hypothesis significance tests). The numerical results of Bayesian and frequentist analysis are often similar. However, the interpretations are very different.

Example 9.1. Recall Example 7.4 which concerned θ, the population proportion of American adults who have read a book in the last year. Recall the actual study data in which 75% of the 1502 American adults surveyed said they read a book in the last year.

We'll compare our Bayesian analysis in Example 7.4 to a frequentist analysis.

1. Compute a 98% frequentist confidence interval for θ.

2. Write a clearly worded sentence reporting the confidence interval in context.

3. Explain what "98% confidence" means.

4. Compare the *numerical results* of the Bayesian and frequentist analysis. Are they similar or different?

5. How does the *interpretation* of these results differ between the two approaches?

6. From a frequentist perspective, which value, 0.73 or 0.75, is more plausible for θ, rounded to two decimal places? Explain.

7. From a Bayesian perspective, which value, 0.73 or 0.75, is more plausible for θ, rounded to two decimal places? Explain.

Example 9.2. Continuing Example 9.1. Have more than 70% of Americans read a book in the last year? We'll now compare a Bayesian analysis to a frequentist (null) hypothesis (significance) test.

Recall the actual study data in which 75% of the 1502 American adults surveyed said they read a book in the last year.

1. Conduct an appropriate hypothesis test.

2. Write a clearly worded sentence reporting the conclusion of the hypothesis test in context.

3. Write a clearly worded sentence interpreting the p-value in context.

4. Now back to the Bayesian analysis of Example 7.4. Compute the posterior probability that θ is less than or equal to 0.70.

5. Compare the *numerical values* of the posterior probability and the p-value. Are they similar or different?

6. How does the *interpretation* of these results differ between the two approaches?

In a Bayesian approach

- Parameters are random variables and have distributions.
- Observed data are treated as fixed, not random.

- All inference is based on the posterior distribution of parameters which quantifies our uncertainty about the parameters.
- The posterior distribution quantifies our uncertainty in the parameters, after observing the sample data.
- The posterior (or prior) distribution can be used to make probability statements about parameters.
- For example, "95% credible" quantifies our assessment that the parameter is 19 times more likely/plausible to lie inside the credible interval than outside. (Roughly, we'd be willing to bet at 19-to-1 odds on whether θ lies inside the interval.)

In a frequentist approach

- Parameters are treated as fixed (not random), but unknown numbers
- Data are treated as random
- All inference is based on the sampling distribution of the data which quantifies how the data behaves over many hypothetical samples.
- For example, "95% confidence" is confidence in the procedure: confidence intervals vary from sample-to-sample; over many samples 95% of confidence intervals contain the parameter being estimated.
- p-values are confusing

Example 9.3. Recall Example 8.4 in which we assumed body temperatures (degrees Fahrenheit) of healthy adults follow a Normal distribution with unknown mean θ and known standard deviation $\sigma = 1$, and our goal was to estimate θ, the population mean healthy human body temperature.

We performed a Bayesian analysis based on a sample of 208 healthy adults with a sample mean body temperature of 97.7 degrees F.

1. Compute a 98% frequentist confidence interval for θ.

2. Write a clearly worded sentence reporting the confidence interval in context.

3. Compare the *numerical results* of the Bayesian and frequentist analysis. Are they similar or different?

4. How does the *interpretation* of these results differ between the two approaches?

5. From a frequentist perspective, which value, 97.6 or 97.7, is more plausible for θ, rounded to one decimal place? Explain.

6. From a Bayesian perspective, which value, 97.6 or 97.7, is more plausible for θ, rounded to one decimal place? Explain.

Example 9.4. Continuing Example 9.3. Is population mean healthy human body temperature less than 98.6 degrees Fahrenheit? We'll now compare a Bayesian analysis to a frequentist (null) hypothesis (significance) test.

1. Conduct an appropriate hypothesis test.

2. Write a clearly worded sentence reporting the conclusion of the hypothesis test in context.

3. Write a clearly worded sentence interpreting the p-value in context.

4. Now back to the Bayesian analysis of Example 8.4. Compute the posterior probability that θ is greater than or equal to 98.6.

5. Compare the *numerical values* of the posterior probability and the p-value. Are they similar or different?

6. How does the *interpretation* of these results differ between the two approaches?

10 Introduction to Prediction

- A Bayesian analysis leads directly and naturally to making predictions about future observations from the random process that generated the data.
- Prediction is also useful for tuning prior distributions, and for checking if model assumptions seem reasonable in light of observed data.

Example 10.1. Do people prefer to use the word "data" as singular or plural? FiveThirtyEight conducted a poll to address this question (and others). Rather than simply ask whether the respondent considered "data" to be singular or plural, they asked which of the following sentences they prefer:

a. Some experts say it's important to drink milk, but the data is inconclusive.
b. Some experts say it's important to drink milk, but the data are inconclusive.

Suppose we wish to study the opinions of students in Cal Poly statistics classes regarding this issue. That is, let θ represent the population proportion of students in Cal Poly statistics classes who prefer to consider data as a *singular* noun, as in option a) above ("the data is").

Before proceeding, consider the context of this problem and sketch your prior distribution for θ. What are the main features of your prior?

To illustrate ideas, we'll start with a prior distribution which places probability 0.01, 0.05, 0.15, 0.30, 0.49 on the values 0.1, 0.3, 0.5, 0.7, 0.9, respectively.

1. Before observing any data, suppose we plan to randomly select a single Cal Poly statistics student. Consider the *unconditional* prior probability that the selected student prefers data as singular. (This is called a *prior predictive probability.*) Explain how you could use simulation to approximate this probability.

2. Compute the prior predictive probability from the previous part.

3. Before observing any data, suppose we plan to randomly select a sample of 35 Cal Poly statistics students. Consider the *unconditional* prior distribution of the number of students in the sample who prefer data as singular. (This is called a *prior predictive distribution*.) Explain how you could use simulation to approximate this distribution. In particular, how could you use simulation to approximate the prior predictive probability that exactly 31 students in the sample prefer data as singular?

4. Compute and interpret the prior predictive probability that exactly 31 students in a sample of size 35 prefer data as singular.

Example 10.2. Continuing Example 10.1. Suppose that we observe a sample of 35 Cal Poly statistics students in which 31 prefer data as singular.

1. Explain how you would use simulation to approximate the posterior distribution of θ.

2. Find the posterior distribution of θ.

Example 10.3. Continuing Example 10.2. After observing a sample of 35 Cal Poly statistics students in which 31 prefer data as singular, suppose we plan to collect more data.

1. Suppose we plan to randomly select an additional Cal Poly statistics student. Consider the *posterior predictive probability* that this student prefers data as singular. Explain how you could use simulation to estimate this probability.

2. Compute the posterior predictive probability from the previous part.

3. Suppose we plan to collect data on another sample of 35 Cal Poly statistics students. Consider the *posterior predictive distribution* of the number of students in the new sample who prefer data as singular. Explain how you could use simulation to approximate this distribution. In particular, how could you use simulation to approximate the posterior predictive probability that exactly 31 students in the sample prefer data as singular? (Of course, the sample size of the new sample does not have to be 35; however, we're keeping it the same so we can compare the prior and posterior predictions.)

4. Compute and interpret the posterior predictive probability that exactly 31 students in a sample of size 35 prefer data as singular.

- The **predictive distribution** of a measured variable y is the marginal distribution (of the unobserved values) after accounting for the uncertainty in the parameters θ.

- A **prior predictive distribution** is calculated using the prior distribution of the parameters.
- A **posterior predictive distribution** is calculated using the posterior distribution of the parameters, conditional on the observed data.
- Be sure to carefully distinguish between posterior distributions and posterior predictive distributions (or between prior distributions and prior predictive distributions.)
  - **Prior and posterior distributions** are distributions on values of the *parameters*. These distributions quantify the degree of uncertainty about unknown parameters θ (before and after observing data).
  - **Prior and posterior *predictive* distributions** are distributions on *potential* values of the *measured variable* (data). Predictive distributions reflect sample-to-sample variability of the sample data, while also accounting for the uncertainty in the parameters.
- Simulation is an effective tool in approximating posterior (or prior) predictive distributions.
  - Step 1: Simulate a value of θ from its posterior distribution (or prior distribution).
  - Step 2: Given this value of θ simulate a value of y from $f(y|\theta)$, the data model conditional on θ.
  - Repeat many times to simulate many (θ, y) pairs, and summarize the values of y to approximate the posterior predictive distribution (or prior predictive distribution).

Example 10.4. Continuing Example 10.1. We'll use a grid approximation and assume that any multiple of 0.0001 is a possible value of θ: 0, 0.0001, 0.0002, ..., 0.9999, 1.

1. Assume the prior distribution for θ is proportional to θ^2. Plot this prior distribution and describe its main features. Is the prior distribution truly discrete or essentially continuous?

2. Given the shape of the prior distribution, explain why we might not want to compute *central* prior credible intervals. Suggest an alternative approach, and compute a 98% prior credible interval for θ.

3. Before observing any data, suppose we plan to randomly select a sample of 35 Cal Poly statistics students. Let y represent the number of students in the selected sample who prefer data as singular. Use simulation to approximate the prior predictive distribution of y and plot it. Is the prior predictive distribution truly discrete or essentially continuous?

4. Find a 95% prior *prediction* interval for y. Write a clearly worded sentence interpreting this interval in context.

Example 10.5. Continuing Example 10.4. Suppose that we observe a sample of 35 Cal Poly statistics students in which 31 prefer data as singular.

1. Find the posterior distribution of θ, plot it and describe its main features, and find and interpret a 98% central posterior credible interval for θ. Is the posterior distribution truly discrete or essentially continuous?

2. Suppose we plan to randomly select another sample of 35 Cal Poly statistics students. Let $\tilde{y}$ represent the number of students in the new sample who prefer data as singular. Use simulation to approximate the posterior predictive distribution of $\tilde{y}$ and plot it. Is the posterior predictive distribution truly discrete or essentially continuous? (Of course, the sample size of the new sample does not have to be 35. However, we're keeping it the same so we can compare the prior and posterior predictions.)

3. Find a 95% posterior *prediction* interval for $\tilde{y}$. Write a clearly worded sentence interpreting this interval in context.

Example 10.6. Continuing Example 10.5. Now suppose instead of using the Cal Poly sample data (31/35) to form the posterior distribution of θ, we had used the data from the FiveThirtyEight study in which 865 out of 1093 respondents preferred data as singular.

1. Find the posterior distribution of θ, plot it and describe its main features, and find and interpret a 98% central posterior credible interval for θ. How does the posterior based on the FiveThirtyEight data compare to the posterior distribution based on the Cal Poly sample data (31/35)? Why?

2. Again, suppose we use the FiveThirtyEight data to form the posterior distribution of θ. Suppose we plan to randomly select a new sample of size 35. Let $\tilde{y}$ represent the number in the new sample who prefer data as singular. Use simulation to approximate the posterior predictive distribution of $\tilde{y}$ and plot it. Find a 95% posterior *prediction* interval for $\tilde{y}$. How does the predictive distribution which uses the posterior distribution based on the FiveThirtyEight data compare to the one based on the Cal Poly sample data (31/35)? Why?

- Be sure to distinguish between a prior/posterior *distribution* and a prior/posterior *predictive distribution*.
- A prior/posterior *distribution* is a distribution on potential values of the *parameters* θ. These distributions quantify the degree of uncertainty about the unknown parameter θ (before and after observing data).

- A prior/posterior *predictive distribution* is a distribution on potential values of the *data* y. Predictive distributions reflect sample-to-sample variability of the sample data, while accounting for the uncertainty in the parameters.
- Even if parameters are essentially "known" — that is, even if the prior/posterior SD of parameters is small — there will still be sample-to-sample variability reflected in the predictive distribution of the data, mainly influenced by the size n of the sample being "predicted" (or unit-to-unit variability for the predictive distribution of a single observation).

Example 10.7. Suppose we want to estimate θ, the population mean hours of sleep on a typical night for Cal Poly students. Assume that sleep hours, y, for students follow a Normal distribution with unknown mean θ and known standard deviation 1.5 hours. (Known population SD is an unrealistic assumption that we use for simplicity here.)

Assume a Normal(7.5, 0.5) prior distribution for θ.

1. Describe what the prior distribution represents. Find and interpret a 98% prior credible interval.

2. Explain how you would use simulation to approximate the prior predictive distribution of y. Use simulation to approximate a 95% prior prediction interval, and interpret it.

3. Now suppose you observe sleep hours (rounded to one decimal place) of 6.8 and 7.2 for a sample of 2 students. Explain how, in principle, you could use simulation to approximate the posterior distribution. (To preview what lies ahead: what issues would you experience with this method in practice?)

4. Suppose that the posterior distribution is approximately Normal with mean 7.4 and SD 0.4. Describe what the posterior distribution represents. Find and interpret a 98% posterior credible interval.

5. Explain how you would use simulation to approximate the posterior predictive distribution of y. Use simulation to approximate a 95% posterior prediction interval, and interpret it.

6. Now suppose you observe a very large sample of students (with a sample mean of 7.0), and the posterior distribution is Normal with mean 7.0 and SD 0.001. Without doing any calculations, approximate the posterior predictive distribution of y and a 95% posterior prediction interval.

10.1 Notes

10.1.1 Prior predictive probability of success for single trial

1. Simulate a value of θ from the prior distribution.
2. Given the value of θ, construct a spinner that lands on success with probability θ. Spin the spinner once and record the result, success or not.
3. Repeat steps 1 and 2 many times, and find the proportion of repetitions which result in success. This proportion approximates the (unconditional) prior predictive probability of success.

```
n_rep = 1000
```

11 Predictive Tuning and Checking

11.1 Prior predictive tuning

- Prior distributions of parameters quantify uncertainty about parameters before observing data.
- Considering prior predictive distributions of possible samples under the proposed model can help tune prior distributions of parameters.

Example 11.1. Suppose we're interested in studying hours of sleep on a typical weeknight for Cal Poly students.

1. What percent of students do you think sleep more than 10 hours on a typical weeknight? Less than 5 hours?

2. Suppose we want to perform a Bayesian analysis to estimate θ, the population mean hours of sleep on a typical weeknight for Cal Poly students. Assume that sleep hours for individual students follow a Normal distribution with unknown mean θ and known standard deviation $\sigma = 1.5$ hours. (Known population SD is an unrealistic assumption that we use for simplicity here.) Suppose we want to "let the data speak for itself" so we choose a flat, Uniform prior[1]. Our Uniform prior needs an upper and lower bound. We remember that we shouldn't assign 0 prior probability to possible values, no matter how implausible, so we pick a fairly wide range of possible values for θ, namely Uniform on [3, 13]. Use simulation to approximate the prior predictive distribution of sleep hours according to our model.

[1] We'll discuss some problems with this approach later.

3. According to this model, (approximately) what percent of students sleep more than 10 hours on a typical weeknight? Less than 5? Do these values seem reasonable? If not, what should we do?

- Prior predictive distributions "live" on the scale of the data, and are sometimes easier to interpret than prior distributions themselves, especially when there are multiple parameters.
- It is often helpful to tune prior distributions of parameters indirectly via prior predictive distributions rather than directly.
- We can choose a prior distribution for parameters, simulate a prior predictive distribution for the data (according to our conditional model) given this prior, and consider if the distribution of possible data values seems reasonable given our background knowledge about the variable and context.
- If not, we can choose another prior distribution and repeat the process until we have suitably "tuned" the prior distribution for the parameters.
- Remember, the prior distribution does not have to be perfect; there is no perfect prior distribution. However, if a particular prior distribution gives rise to obviously unreasonable data values (e.g., negative sleep hours) we should try to improve it. It's always a good idea to consider prior predictive distributions when formulating a prior distribution for parameters.
- Also remember, there are many assumptions that go into any model. If the prior predictive distribution indicates potential inadequacy of the model, it is not necessarily the fault of the prior distribution.

11.2 Posterior predictive checking

- Posterior predictive distributions can be used to check if model assumptions seem reasonable in light of observed data.

Example 11.2. Recall Example 10.1 which concerned θ, the population proportion of students in Cal Poly statistics classes who prefer to consider data as a singular noun (as in "the data is"). Suppose that before collecting data for our sample of Cal Poly students, we had based our

prior distribution off the FiveThirtyEight data. In particular, suppose we assume a Normal prior distribution for θ with prior mean 0.792 and prior SD 0.012.

1. What does this our prior distribution say about our beliefs about θ?

2. Now suppose we randomly select a sample of 35 Cal Poly students and 21 students prefer data as singular. Find the posterior distribution, and plot prior, likelihood, and posterior. Have our beliefs about θ changed? Why?

3. Use simulation to find the posterior predictive distribution corresponding to samples of size 35. Compare the observed sample value of 21/35 with the posterior predictive distribution. What do you notice? Does this indicate problems with the model?

- A Bayesian model is composed of both a model for the data (likelihood) and a prior distribution on model parameters.
- Predictive distributions can be used as tools in model checking.
- **Posterior predictive checking** involves comparing the observed data to simulated samples (or some summary statistics) generated from the posterior predictive distribution.
- We'll focus on graphical checks: Compare plots for the observed data with those for simulated samples. Systematic differences between simulated samples and observed data indicate potential shortcomings of the model.
- If the model fits the data, then replicated data generated under the model should look similar to the observed data.
- If the observed data is not plausible under the posterior predictive distribution, then this could indicate that the model is not a good fit for the data. ("Based on the data we observed, we conclude that it would be unlikely to observe the data we observed???")
- However, a problematic model isn't necessarily due to the prior. Remember that a Bayesian model consists of both a prior and a likelihood, so model mis-specification

can occur in the prior or likelihood or both. The form of the likelihood is also based on subjective assumptions about the variables being measured and how the data are collected. Posterior predictive checking can help assess whether these assumptions are reasonable in light of the observed data.

Example 11.3. A basketball player will attempt a sequence of 20 free throws. Our model assumes

- The probability that the player successfully makes any particular free throw attempt is θ.
- A Uniform prior distribution for θ (in a grid from 0 to 1).
- Conditional on θ, the number of successfully made attempts has a Binomial(20, θ) distribution. (This determines the likelihood.)

1. Suppose the player misses her first 10 attempts and makes her second 10 attempts. Does this data seem consistent with the model?

2. Explain how you could use posterior predictive checking to check the fit of the model.

11.3 Notes

11.3.1 Prior predictive distribution of sleep hours

```
n_rep = 10000

# simulate theta from prior
theta_sim = runif(n_rep, 3, 13)

# given theta, simulation y from model for data
sigma = 1.5
```

12 Beta-Binomial Model

- Bayesian analysis is based on the posterior distribution of parameters θ given data y.
- The data y might be discrete (e.g., count data) or continuous (e.g., measurement data).
- However, *parameters* θ almost always take values on a *continuous* scale, even when the data are discrete.
- Now we introduce continuous prior and posterior distributions to quantify uncertainty about parameters. Some general notation:
  - θ represents[1] parameters of interest usually taking values on a continuous scale
  - y denotes observed sample data[2] (discrete or continuous)
  - $\pi(\theta)$ denotes the prior distribution of θ, usually a probability density function (pdf) over possible values of θ
  - $f(y|\theta)$ denotes the likelihood function, a function of continuous θ for fixed y
  - $\pi(\theta|y)$ denotes the posterior distribution of θ, the conditional distribution of θ given the data y.
- Bayes rule works analogously for a continuous parameter θ, given data y

$$\pi(\theta|y) = \frac{f(y|\theta)\pi(\theta)}{f(y)}$$

$$\pi(\theta|y) \propto f(y|\theta)\pi(\theta)$$

$$\text{posterior} \propto \text{likelihood} \times \text{prior}$$

- Predictive distribution can be found using a continuous analog of the law of total probability

$$\text{Prior predictive} \qquad f(y) = \int_{-\infty}^{\infty} f(y|\theta)\pi(\theta)d\theta,$$

$$\text{Posterior predictive} \qquad f(\tilde{y}|\theta) = \int_{-\infty}^{\infty} f(\tilde{y}|\theta)\pi(\theta|y)d\theta.$$

However, we almost always use simulation to approximate predictive distributions.

[1] θ is used to denote both: (1) the actual parameter (i.e., the random variable) θ itself, and (2) possible values of θ.

[2] When considering the prior predictive distribution y also denotes potential values of the data. For the posterior predictive distribution $\tilde{y}$ denotes potential new values of the data.

Example 12.1. Recall Example 10.4 which concerned θ, the population proportion of students in Cal Poly statistics classes who prefer to consider data as a singular noun (as in "the data is").

1. Why might we not want to assume a Normal prior distribution for θ?

2. In Example 10.4 we assumed a prior distribution for θ which is proportional to $\theta^2,\ 0 < \theta < 1$. In that example, we used a grid approximation for the prior distribution, but now we'll treat the distribution as continuous. Sketch this prior distribution.

3. Now we'll consider a few more prior distributions. Sketch each of the following priors. How do they compare?

   a. proportional to $\theta^2,\ 0 < \theta < 1$. (from previous)
   b. proportional to $\theta^5,\ 0 < \theta < 1$.
   c. proportional to $(1-\theta)^2,\ 0 < \theta < 1$.
   d. proportional to $\theta^2(1-\theta)^2,\ 0 < \theta < 1$.
   e. proportional to $\theta^5(1-\theta)^2,\ 0 < \theta < 1$.

- A continuous random variable U has a **Beta distribution** with *shape parameters* $\alpha > 0$

and $\beta > 0$ if its density (pdf) satisfies[3]

$$p(u) \propto u^{\alpha-1}(1-u)^{\beta-1}, \quad 0 < u < 1.$$

- If $\alpha = \beta$ the distribution is symmetric about 0.5
- If $\alpha > \beta$ the distribution is skewed to the left (with greater density above 0.5 than below)
- If $\alpha < \beta$ the distribution is skewed to the right (with greater density below 0.5 than above)
- If $\alpha = 1$ and $\beta = 1$, the Beta(1, 1) distribution is the Uniform distribution on (0, 1).
- It can be shown that a Beta(α, β) distribution has

$$\begin{aligned} \text{Mean (EV):} \quad & \frac{\alpha}{\alpha+\beta} \\ \text{Variance:} \quad & \frac{\left(\frac{\alpha}{\alpha+\beta}\right)\left(1-\frac{\alpha}{\alpha+\beta}\right)}{\alpha+\beta+1} \\ \text{Mode:} \quad & \frac{\alpha-1}{\alpha+\beta-2}, \qquad \text{(if } \alpha > 1,\ \beta \geq 1 \text{ or } \alpha \geq 1,\ \beta > 1\text{; otherwise, there is not a unique mode)} \end{aligned}$$

Example 12.2. Continuing Example 12.1.

(6) Each of the distributions in the previous example was a Beta distribution. For each distribution, identify the shape parameters and the prior mean and standard deviation.

Prior proportional to	Prior α	Prior β	Prior Mean	Prior SD
θ^2				
θ^5				
$(1-\theta)^2$				
$\theta^2(1-\theta)^2$				
$\theta^5(1-\theta)^2$				

(7) Assume a Beta(3, 1) prior distribution for θ. Now suppose that 31 students in a sample of 35 Cal Poly statistics students prefer data as singular. Write the function that defines the shape of the likelihood as a function of $\theta, 0 < \theta < 1$.

[3]The expression defines the shape of the Beta density. All that's missing is the scaling constant which ensures that the total area under the density is 1. The actual Beta density formula, including the normalizing constant, is

$$p(u) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\, u^{\alpha-1}(1-u)^{\beta-1}, \quad 0 < u < 1,$$

where $\Gamma(\alpha) = \int_0^\infty e^{-v} v^{\alpha-1} dv$ is the *Gamma function*. For a positive integer k, $\Gamma(k) = (k-1)!$. Also, $\Gamma(1/2) = \sqrt{\pi}$.

(8) Write the function that defines the shape of the posterior density as a function of $\theta, 0 < \theta < 1$. Is this a Beta distribution? If so, specify the values of α and β.

(9) Starting with each of the prior distributions from the first part, find the posterior distribution of θ based on this sample, and identify it as a Beta distribution by specifying the shape parameters α and β.

Prior	Posterior α	Posterior β	Posterior Mean	Posterior SD
Beta(3, 1)				
Beta(6, 1)				
Beta(1, 3)				
Beta(3, 3)				
Beta(6, 3)				

(10) For each of the posterior distributions in the previous part, compute the posterior mean and standard deviation. How does each posterior distribution compare to its respective prior distribution?

- **Beta-Binomial model.** If θ has a Beta(α, β) prior distribution and the conditional distribution of y given θ is the Binomial(n, θ) distribution, then the posterior distribution

of θ given y is the Beta$(\alpha + y, \beta + n - y)$ distribution[4].

$$\text{prior:} \qquad \pi(\theta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}, \quad 0 < \theta < 1,$$

$$\text{likelihood:} \qquad f(y|\theta) \propto \theta^{y}(1-\theta)^{n-y}, \quad 0 < \theta < 1,$$

$$\text{posterior:} \qquad \pi(\theta|y) \propto \theta^{\alpha+y-1}(1-\theta)^{\beta+n-y-1}, \quad 0 < \theta < 1.$$

- In a sense, you can interpret α as "prior successes" and β as "prior failures", but these are only "pseudo-observations". Also, α and β are not necessarily integers.

	Prior	Data	Posterior
Successes	α	y	$\alpha + y$
Failures	β	$n - y$	$\beta + n - y$
Total	$\alpha + \beta$	n	$\alpha + \beta + n$

Example 12.3. In Example 10.4 we essentially assumed a Beta(3, 1) prior but we used grid approximation to compute the posterior. Now we'll compare our results from that exercise to those using the continuous Beta(3, 1) prior.

Assume that θ has a Beta(3, 1) prior distribution and that 31 students in a sample of 35 Cal Poly statistics students prefer data as singular.

1. Plot the prior distribution, (scaled) likelihood, and posterior distribution.

2. Use software to find 50%, 80%, and 98% central posterior credible intervals.

[4]When the prior and posterior distribution belong to the same family, that family is called a **conjugate** prior distribution for the likelihood. So, the Beta distributions form a conjugate prior family for Binomial distributions.

3. Compare the results to those using the grid approximation.

4. Express the posterior mean as a weighted average of the prior mean and sample proportion. Describe what the weights are, and explain why they make sense.

- In the Beta-Binomial model, the posterior mean $E(\theta|y)$ can be expressed as a *weighted average* of the prior mean $E(\theta) = \frac{\alpha}{\alpha+\beta}$ and the sample proportion $\hat{p} = y/n$.

$$E(\theta|y) = \frac{\alpha+\beta}{\alpha+\beta+n}E(\theta) + \frac{n}{\alpha+\beta+n}\hat{p}$$

- As more data are collected, more weight is given to the sample proportion (and less weight to the prior mean).
- The prior "weight" is determined by $\alpha+\beta$, which is sometimes called the *concentration* and measured in "pseudo-observations".
- Larger values of $\alpha+\beta$ indicate stronger prior beliefs, due to smaller prior variance, and give more weight to the prior mean.
- The posterior variance generally gets smaller as more data are collected

$$\text{Var}(\theta|y) = \frac{E(\theta|y)(1-E(\theta|y))}{\alpha+\beta+n+1}$$

Example 12.4. Continuing the previous examples, we'll now consider predictive distributions. Assume that θ has a Beta(3, 1) prior distribution.

1. Before observing any data, suppose we plan to randomly select a sample of 35 Cal Poly statistics students. Let y represent the number of students in the selected sample who prefer data as singular. Explain how we could use simulation to approximate the prior predictive distribution of y. Then conduct the simulation and approximate the prior predictive distribution. Is the prior predictive distribution truly discrete or continuous? How does the prior predictive distribution compare to the one from Example 10.4?

2. Now suppose we observe 31 students in a sample of 35 Cal Poly statistics students prefer data as singular, so that the posterior distribution of θ is the Beta(34, 5) distribution. Suppose we plan to randomly select another sample of 35 Cal Poly statistics students. Let $\tilde{y}$ represent the number of students in the selected sample who prefer data as singular. Explain how we could use simulation to approximate the posterior predictive distribution of $\tilde{y}$. Then conduct the simulation and approximate the posterior predictive distribution. Is the posterior predictive distribution truly discrete or continuous? How does the posterior predictive distribution compare to the one from Example 10.5?

3. Use the simulation results to approximate a 95% posterior *prediction* interval for $\tilde{y}$. Write a clearly worded sentence interpreting this interval in context.

- You can tune the shape parameters — α (like "prior successes") and β (like "prior failures") — of a Beta distribution to your prior beliefs in a few ways. Recall that $\kappa = \alpha + \beta$ is the "concentration" or "equivalent prior sample size".
- If prior mean μ and prior concentration κ are specified then

$$\begin{aligned}\alpha &= \mu\kappa \\ \beta &= (1-\mu)\kappa\end{aligned}$$

- If prior mode ω and prior concentration κ (with $\kappa > 2$) are specified then

$$\begin{aligned}\alpha &= \omega(\kappa-2)+1 \\ \beta &= (1-\omega)(\kappa-2)+1\end{aligned}$$

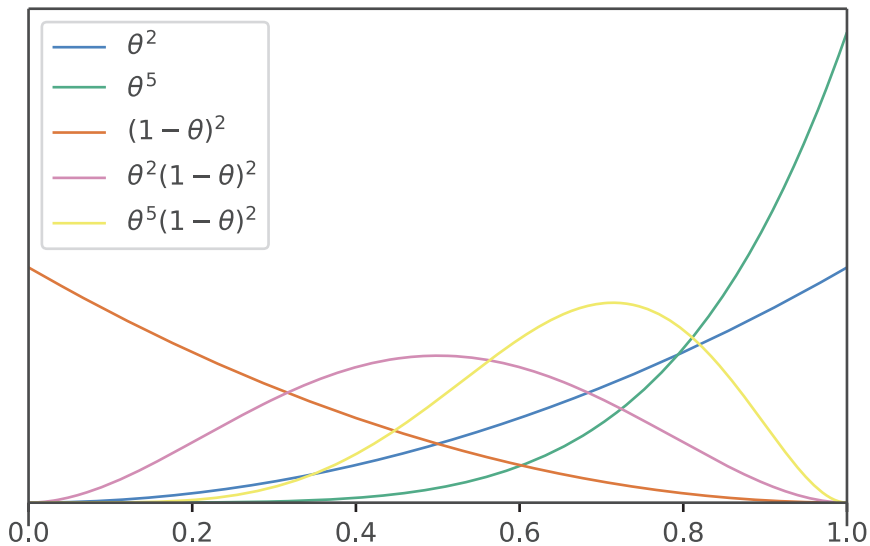
- If prior mean μ and prior sd σ are specified then

$$\begin{aligned}\alpha &= \mu\left(\frac{\mu(1-\mu)}{\sigma^2}-1\right) \\ \beta &= (1-\mu)\left(\frac{\mu(1-\mu)}{\sigma^2}-1\right)\end{aligned}$$

- You can also specify two percentiles and use software to find α and β. For example, you could specify the endpoints of a prior 98% credible interval.

12.1 Notes

12.1.1 Prior Beta distributions



	Distribution	α	β	Proportional to	Mean	SD
a	Beta(3, 1)	3	1	$\theta^2, 0 < \theta < 1$	0.750	0.194
b	Beta(6, 1)	6	1	$\theta^5, 0 < \theta < 1$	0.857	0.124
c	Beta(1, 3)	1	3	$(1-\theta)^2, 0 < \theta < 1$	0.250	0.194
d	Beta(3, 3)	3	3	$\theta^2(1-\theta)^2, 0 < \theta < 1$	0.500	0.189
e	Beta(6, 3)	6	3	$\theta^5(1-\theta)^2, 0 < \theta < 1$	0.667	0.149

13 Considering Prior Distributions

One of the most commonly asked questions when one first encounters Bayesian statistics is "how do we choose a prior?" While there is never one "perfect" prior in any situation, we'll noew discuss some issues to consider when choosing a prior. But first, here are a few big picture ideas to keep in mind:

- Bayesian inference is based on the posterior distribution, not the prior. Therefore, the posterior requires much more attention than the prior.
- The prior is only one part of the Bayesian model.
- In many situations, the posterior distribution is not too sensitive to reasonable changes in prior.
- Any statistical analysis is inherently subjective. Priors and Bayesian data analysis are no more inherently subjective than any of the myriad other assumptions made in statistical analysis.
- Both Bayesian and frequentist are valid approaches to statistical analyses, each with advantages and disadvantages.
- There are some issues with frequentist approaches that incorporating a prior distribution and adopting a Bayesian approach alleviates.

Example 13.1. Tamika is a basketball player who throughout her career has had a probability of 0.5 of making any three point attempt. However, her coach is afraid that her three point shooting has gotten worse. To check this, the coach has Tamika shoot a series of three pointers; she makes 7 out of 24. Does the coach have evidence that Tamika has gotten worse?

Let θ be the probability that Tamika successfully makes any three point attempt. Assume attempts are independent.

1. Prior to collecting data, the coach decides that he'll have convincing evidence that Tamika has gotten worse if the p-value is less than 0.025. Suppose the coach told Tamika to *shoot 24 attempts and then stop* and count the number of successful attempts. Use software to compute the p-value. Is the coach convinced that Tamika has gotten worse?

2. Prior to collecting data, the coach decides that he'll have convincing evidence that Tamika has gotten worse if the p-value is less than 0.025. Suppose the coach told Tamika to *shoot until she makes 7 three pointers and then stop* and count the number of total attempts. Use software to compute the p-value. Is the coach convinced that Tamika has gotten worse? (Hint: the total number of attempts has a Negative Binomial distribution.)

3. Now suppose the coach takes a Bayesian approach and assumes a Beta(α, β) prior distribution for θ. Suppose the coach told Tamika to *shoot 24 attempts and then stop* and count the number of successful attempts. Identify the likelihood function and the posterior distribution of θ.

4. Now suppose the coach takes a Bayesian approach and assumes a Beta(α, β) prior distribution for θ. Suppose the coach told Tamika to *shoot until she makes 7 three pointers and then stop* and count the number of total attempts. Identify the likelihood function and the posterior distribution of θ.

5. Compare the Bayesian and frequentist approaches in this example. Does the "strength of the evidence" depend on how the data were collected?

- Some issues to consider when choosing a prior include, in no particular order:

- *The researcher's prior beliefs!* A prior distribution is part of a statistical model, and should be consistent with knowledge about the underlying scientific problem. Researchers are often experts with a wealth of past experience that can be explicitly incorporated into the analysis via the prior distribution. Such a prior is called an informative (or weakly informative) prior.
- *A regularizing prior.* A prior which, when tuned properly, reduces overfitting or "over-reacting" to the data.
- *Noninformative prior a.k.a., (reference, vague, flat prior).* A prior is sought that plays a minimal role in inference so that "the data can speak for itself".
- *Mathematical convenience.* The prior is chosen so that computation of the posterior is simplified, as in the case of conjugate priors.
- *Interpretation.* The posterior is a compromise between the data and prior. Some priors allow for easy interpretation of the relative contributions of data and prior to the posterior. For example, think of the "prior successes and prior failures" interpretation in the Beta-Binomial model.
- *Prior based on* past *data.* Bayesian updating can be viewed as an iterative process. The posterior distribution obtained from one round of data collection can inform the prior distribution for another round.
- *Prior predictive distributions.* Prior predictive distributions of potential data can be used to check the reasonableness of a prior distribution of parameters before observing sample data. Prior predictive distributions "live" on the scale of the data, and are sometimes easier to interpret than prior distributions themselves, especially when there are multiple parameters. It is often helpful to tune prior distributions of parameters indirectly via prior predictive distributions rather than directly.

Example 13.2. Suppose we want to estimate θ, the population proportion of Cal Poly students who wore socks at any point yesterday.

1. What are the possible values for θ? What prior distribution might you consider a noninformative prior distribution?

2. You might choose a Uniform(0, 1) prior, a.k.a., a Beta(1, 1) prior. Recall how we interpreted the parameters α and β in the Beta-Binomial model. Does the Beta(1, 1) distribution represent "no prior information"?

3. Suppose in a sample of 20 students, 4 wore socks yesterday. How would you estimate θ with a single number based only on the data?

4. Assume a Beta(1, 1) prior and the 4/20 sample data. Identify the posterior distribution. Recall that one Bayesian point estimate of θ is the posterior mean. Find the posterior mean of θ. Does this estimate let the "data speak entirely for itself"?

5. How could you change α and β in the Beta distribution prior to represent no prior information? Sketch the prior. Do you see any potential problems?

6. Assume a Beta(0, 0) prior for θ and the 4/20 sample data. Identify the posterior distribution. Find the posterior *mode* of θ. Does this estimate let the "data speak entirely for itself"?

7. Now suppose the parameter you want to estimate is the *odds* that a student wore socks yesterday, $\phi = \frac{\theta}{1-\theta}$. What are the possible values of ϕ? What might a non-informative prior look like? Is this a proper prior?

8. Assume a Beta(1, 1) prior for θ. Use simulation to approximate the prior distribution of the odds ϕ. Would you say this is a noninformative prior for ϕ?

- An *improper* prior distribution is a prior distribution that does not integrate to 1, so is not a proper probability density.
- However, an improper proper often results in a proper posterior distribution.
- Flat priors are common choices in some situations, but are rarely ever the best choices from a modeling perspective.
- Furthermore, flat priors are generally not preserved under transformations of parameters.

Example 13.3. Suppose that θ represents the population proportion of adults who have a particular rare disease.

1. Explain why you might not want to use a flat Uniform(0, 1) prior for θ.

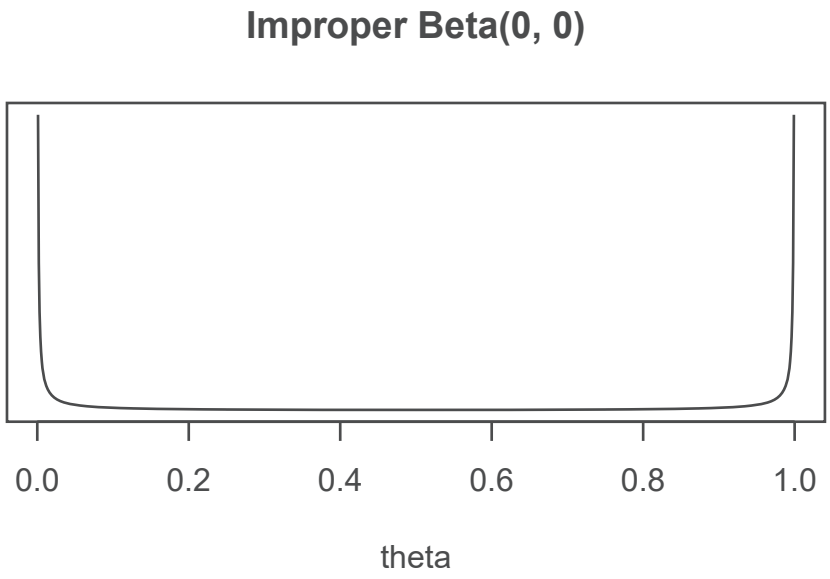
2. Assume a Uniform(0, 1) prior. Suppose you will test $n = 100$ suspected cases. Use simulation to approximate the prior predictive distribution of the number in the sample who have the disease. Does this seem reasonable?

3. Assume a Uniform(0, 1) prior. Suppose that in $n = 100$ suspected cases, none actually has the disease. Find and interpret the posterior median. Does this seem reasonable?

- **What NOT to do when considering priors.**
- Do NOT choose a prior that assigns 0 probability/density to *possible* values of the parameter regardless of how initially implausible the values are.
- Do NOT base the *prior* on the observed data you will use to compute the posterior.
- Do NOT feel like you have to find that one, perfect prior.
- Do NOT worry too much about the prior!

13.1 Notes

13.1.1 Improper Beta(0, 0) prior



13.1.2 Prior distribution of odds

1. Simulate a value of θ from the Beta(1, 1) prior distribution.
2. Compute the odds $\phi = \theta/(1-\theta)$.
3. Repeat many times and summarize the simulated ϕ values to approximate the prior distribution of ϕ.

14 Introduction to Posterior Simulation

- All of Bayesian inference is based on the posterior distribution of parameters given sample data.
- Therefore, the main computational task in a Bayesian analysis is computing, or approximating, the posterior distribution.

Example 14.1. Suppose we wish to estimate the proportion of Cal Poly students that are left-handed. Suppose we want to assume a Normal distribution prior for θ with mean 0.15 and SD 0.08. Also suppose that in a sample of 25 Cal Poly students 5 are left-handed. We want to find the posterior distribution.

Note: the Normal distribution prior assigns positive (but small) density outside of (0, 1). So we can either truncate the prior to 0 outside of (0, 1) or just rely on the fact that the likelihood will be 0 for θ outside of (0, 1) to assign 0 posterior density outside (0, 1).

1. Write an expression for the shape of the posterior density. Is this a recognizable probability distribution?

2. Describe in full detail two methods for approximating the posterior distribution in this situation.

- In most problems it is not possible to find the posterior distribution analytically, and therefore we must approximate it.
- We have seen two approximation methods: grid approximation and simulation.

Example 14.2. Suppose you're interested in studying sleep hours of Cal Poly undergraduates. You assume that within each of the six colleges sleep hours follow a Normal distribution with a mean and SD for the college. You'll collect data on a random sample of students, record their college and sleep hours, and use it to perform a Bayesian analysis.

1. How many parameters are there in this situation?

2. Suppose you try a grid approximation. You create a grid of 100 values over the possible range of values of each parameter (say [3, 13]). How many rows would there be in the Bayes table?

- Grid approximation suffers from the "curse of dimensionality" and is not computationally feasible in multi-parameter problems (and most interesting problems are multi-parameter).

Example 14.3. Recall that we have used simulation to approximate posterior distributions in a few situations: Example 2.3, Example 10.2, Example 10.7.

1. Explain the general method for using simulation to approximate a posterior distribution given sample data. Careful: don't confuse posterior distributions with posterior predictive distributions.

2. Explain why this method is typically not feasible in practice. (It might help to look at the simulation results from the examples referenced above.)

- We have seen three ways to compute/approximate the posterior distribution
  - *Math.* Only works in a few special cases (e.g., Beta-Binomial, Normal-Normal, conjugate prior situations).
  - *Grid approximation.* Not feasible when there are more than just a few parameters.
  - *Simulation.* By far the most common method, though we have only seen a very naive approach so far.
- In principle, the posterior distribution $\pi(\theta|y)$ of parameters θ given observed data y can be approximated via simulation in the following way.
  - Simulate a θ value from the prior distribution (θ could be a vector of parameters, e.g., $\theta = (\mu, \sigma)$).
  - Given the simulated θ, simulate hypothetical data y from the corresponding conditional distribution of y given θ (y could be a full sample, or the value of a sample statistic).
  - Repeat many times and discard (θ, y) pairs for which the simulated data y is not equal to the observed data y
  - Summarize the simulated θ values for the remaining pairs (with y equal to the observed data).
- While this method works in principle, it is a very computationally inefficient way of approximating the posterior distribution, since in most cases (e.g., not-small sample, continuous variables) the probability of replicating the observed data y is essentially 0.
- Therefore, we need more efficient simulation algorithms for approximating posterior distributions. **Markov chain Monte Carlo (MCMC)** methods provide powerful and widely applicable algorithms for simulating from probability distributions, including complex and high-dimensional distributions. These algorithms include Metropolis-Hastings, Gibbs sampling, Hamiltonian Monte Carlo (HMC), and No U-Turn sampling (NUTS), among others. We will see later some of the ideas behind MCMC algorithms.
- Regardless of which simulation algorithm we use to approximate the posterior distribution, there are 3 main inputs:
  - Observed data y
  - Model for the data, $f(y|\theta)$ which depends on parameters θ. (This model determines the likelihood function.)
  - Prior distribution for parameters $\pi(\theta)$
- We then employ some simulation algorithm to approximate the posterior distribution of θ given the observed data y, $\pi(\theta|y)$, *without computing the posterior distribution.*

15 Bayesian Analysis of Poisson Count Data

- We have covered in some detail the problem of estimating a population proportion for a binary categorical variable. In these situations we assumed a Binomial likelihood for the count of "successes" in the sample.
- However, a Binomial model has several restrictive assumptions that might not be satisfied in practice.
- Poisson models are more flexible models for count data.

Example 15.1. Let y be the number of home runs hit (in total by both teams) in a randomly selected Major League Baseball game.

1. In what ways is this like the Binomial situation? (What is a trial? What is "success"?)

2. In what ways is this NOT like the Binomial situation?

- Poisson distributions are often used to model the distribution of variables that count the number of "relatively rare" events that occur over a certain interval of time or in a certain location
- A discrete random variable Y has a **Poisson distribution** with parameter $\theta > 0$ if its probability mass function satisfies

$$f(y|\theta) = \frac{e^{-\theta}\theta^y}{y!}, \quad y = 0, 1, 2, ...$$

- If Y has a Poisson(θ) distribution then

$$\begin{aligned} \mathrm{E}(Y) &= \theta \\ \mathrm{Var}(Y) &= \theta \end{aligned}$$

- **Poisson aggregation.** If Y_1 and Y_2 are independent, Y_1 has a Poisson(θ_1) distribution, and Y_2 has a Poisson(θ_2) distribution, then $Y_1 + Y_2$ has a Poisson($\theta_1 + \theta_2$) distribution.

Example 15.2. Suppose the number of home runs hit per game (by both teams in total) at a particular Major League Baseball park follows a Poisson distribution with parameter θ.

1. Sketch your prior distribution for θ and describe its features. What are the possible values of θ? Does θ take values on a discrete or continuous scale?

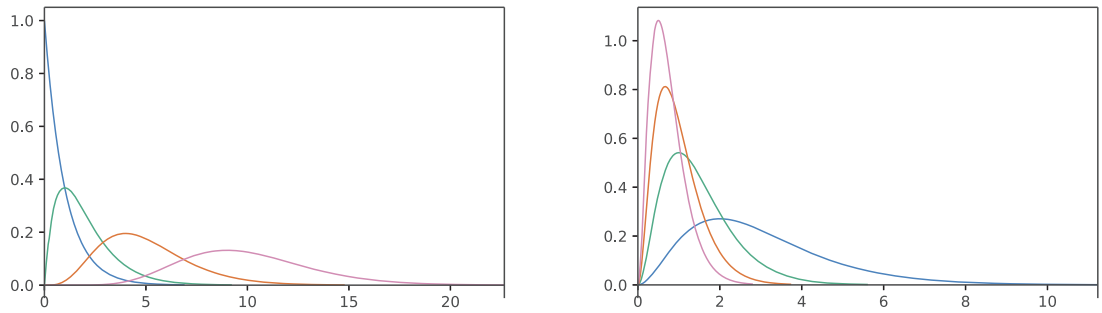
2. Suppose y represents a home run count for a single game. What are the possible values of y? Does y take values on a discrete or continuous scale?

- Gamma distributions are can be used as prior distributions for parameters that take positive values, $\theta > 0$.
- A continuous RV U has a **Gamma distribution** with *shape parameter* $\alpha > 0$ and *rate parameter* $\lambda > 0$ if its density satisfies

$$f(u) \propto u^{\alpha-1} e^{-\lambda u}, \quad u > 0,$$

- It can be shown that a Gamma(α, λ) density has

$$\begin{aligned} \text{Mean (EV)} &= \frac{\alpha}{\lambda} \\ \text{Variance} &= \frac{\alpha}{\lambda^2} \\ \text{Mode} &= \frac{\alpha - 1}{\lambda}, \qquad \text{if } \alpha \geq 1 \end{aligned}$$



(a) Rate parameter 1, and different shape parameters.

(b) Shape parameter 3, and different rate parameters.

Figure 15.1: Gamma densities

Example 15.3. Figure 15.1 shows a few examples of Gamma distributions.

1. The plot on the left above contains a few different Gamma densities, all with rate parameter $\lambda = 1$. Match each density to its shape parameter α; the choices are 1, 2, 5, 10.

2. The plot on the right above contains a few different Gamma densities, all with shape parameter $\alpha = 3$. Match each density to its rate parameter λ; the choices are 1, 2, 3, 4.

Example 15.4. Suppose the number of home runs hit per game (by both teams in total) at a particular Major League Baseball park follows a Poisson distribution with parameter θ. We'll first use grid approximation to approximate the posterior distribution of θ given sample data. We'll start with a discrete prior for θ to illustrate ideas. Suppose that θ can only take values 0.5, 1.5, 2.5, 3.5, 4.5, with respective probabilities 0.13, 0.45, 0.28, 0.11, 0.03.

1. Suppose a single game with 3 home runs is observed. Find the posterior distribution of θ. In particular, how do you determine the likelihood column?

2. Suppose that we observe 4 games with a total of 6 home runs. Find the posterior distribution of θ. In particular, how do you determine the likelihood column?

Example 15.5. Now let's consider some real data. Assume home runs per game at Citizens Bank Park (Phillies!) follow a Poisson distribution with parameter θ. Assume as a prior for θ a Gamma distribution with shape parameter $\alpha = 4$ and rate parameter $\lambda = 2$.

Section 15.1.3 summarizes data for the 2020 season[1]. There were 97 home runs in 32 games.

Use grid approximation to compute the posterior distribution of θ. Plot the prior, (scaled) likelihood, and posterior. (Note: you will need to cut the grid off at some point. While θ can take any value greater than 0, the interval [0, 8] accounts for 99.99% of the prior probability.)

Example 15.6. Continuing Example 15.5.

1. Write an expression for the prior distribution of θ.

[1]Source: https://www.baseball-reference.com/teams/PHI/2020.shtml

2. Write an expression for the likelihood function given the sample data (97 home runs in 32 games).

3. Write an expression for the posterior distribution of θ given the sample data (97 home runs in 32 games). Identify by the name the posterior distribution and the values of relevant parameters. Compare to the results of the grid approximation.

4. Find and interpret posterior 50%, 80%, and 98% credible intervals for θ.

5. Express the posterior mean as a weighted average of the prior mean and the sample mean. How might you interpret the α and β parameters in the Gamma prior distribution?

- **Gamma-Poisson model.** Consider a measured variable y which, given θ, follows a Poisson(θ) distribution. Let $\bar{y}$ be the sample mean for a random sample of size n. Suppose θ has a Gamma(α, λ) prior distribution. Then the posterior distribution of θ given $\bar{y}$ is the Gamma($\alpha + n\bar{y}, \lambda + n$) distribution.
- In a sense, you can interpret α as "prior total count" and λ as "prior sample size", but these are only "pseudo-observations". Also, α and λ are not necessarily integers.
- Note that if $\bar{y}$ is the sample mean count is then $n\bar{y} = \sum_{i=1}^{n} y_i$ is the sample total count.

	Prior	Data	Posterior
Total count	α	$n\bar{y}$	$\alpha + n\bar{y}$
Sample size	λ	n	$\lambda + n$
Mean	$\frac{\alpha}{\lambda}$	$\bar{y}$	$\frac{\alpha+n\bar{y}}{\lambda+n}$

- The posterior total count is the sum of the "prior total count" α and the sample total count $n\bar{y}$.
- The posterior sample size is the sum of the "prior sample size" λ and the observed sample size n.
- The posterior mean is a weighted average of the prior mean and the sample mean, with weights proportional to the "sample sizes".

$$\frac{\alpha + n\bar{y}}{\lambda + n} = \frac{\lambda}{\lambda + n}\left(\frac{\alpha}{\lambda}\right) + \frac{n}{\lambda + n}\bar{y}$$

Example 15.7. Continuing Example 15.5. Continuing the previous example, assume home runs per game at Citizens Bank Park follow a Poisson distribution with parameter θ. Assume for θ a Gamma prior distribution with shape parameter $\alpha = 4$ and rate parameter $\lambda = 2$. Consider the 2020 data in which there were 97 home runs in 32 games.

1. Describe in full detail how you could use naive simulation to approximate the posterior distribution of θ.

2. Write `brms` code to approximate the posterior distribution. Run the code and compare to the previous results.

Example 15.8. Continuing Example 15.5.

1. How could you use simulation to approximate the posterior predictive distribution of home runs in a game?

2. Use the simulation from the previous part to find and interpret a 95% posterior prediction interval.

3. Is a Poisson model a reasonable model for the data? How could you use posterior predictive simulation to simulate what a sample of 32 games might look like under this model. Simulate many such samples. Does the observed sample seem consistent with the model?

4. Regarding the appropriateness of a Poisson model, we might be concerned that there are no games in the sample with 0 home runs. Use simulation to approximate the posterior predictive distribution of the number of games in a sample of 32 with 0 home runs. From this perspective, does the observed value of the statistic seem consistent with the Gamma-Poisson model?

16 Normal-Normal Model

We have estimated the population mean of a Normal distribution, assuming known population standard deviation σ, using grid approximation in Chapter 8 and posterior simulation in Section 14.1. We will relax the unreasonable assumption of known standard deviation soon. But before doing that, we will briefly investigate one more case where the posterior distribution can be derived analytically, in a form that makes explicit the relative influence of prior and likelihood.

Normal-Normal model[1]

- Consider a measured variable y which, given θ, follows a Normal(θ, σ) distribution, with σ known.
- Suppose θ has a Normal(μ_0, τ_0) prior distribution.
- Let $\bar{y}$ be the sample mean for a random sample of size n.
- Then the posterior distribution of θ given $\bar{y}$ is the Normal(μ_n, τ_n) distribution, where

$$\text{posterior precision:} \qquad \frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}$$

$$\text{posterior mean:} \qquad \mu_n = \left(\frac{\frac{1}{\tau_0^2}}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}} \right) \mu_0 + \left(\frac{\frac{n}{\sigma^2}}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}} \right) \bar{y}$$

- The **precision** of a Normal distribution is the reciprocal of its variance.
- If y follows a Normal(θ, σ) distribution, the precision in a single measurement of y is $1/\sigma^2$.
- For a random sample of n values of the variable y, the sample mean $\bar{y}$ follows a Normal distribution with standard deviation $\sigma/\sqrt{n}$ and precision n/σ^2. That is, the precision in n independent data values is n times the precision in a single value.
- For the posterior distribution in the Normal-Normal model, there is an intuitive interpretation of the compromise between prior and likelihood.
  - The posterior precision is the sum of the prior precision and the data precision.
  - The posterior mean is a weighted average of the prior mean and the sample mean, with weights proportional to the precisions.

[1] Try this applet which illustrates the Normal-Normal model.

	Prior	Data (Sample Mean)	Posterior
Precision	$\frac{1}{\tau_0^2}$	$\frac{n}{\sigma^2}$	$\frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}$
SD	τ_0	$\frac{\sigma}{\sqrt{n}}$	τ_n
Mean	μ_0	$\bar{y}$	$\mu_n = \left(\frac{\frac{1}{\tau_0^2}}{\frac{1}{\tau_0^2}+\frac{n}{\sigma^2}}\right)\mu_0 + \left(\frac{\frac{n}{\sigma^2}}{\frac{1}{\tau_0^2}+\frac{n}{\sigma^2}}\right)\bar{y}$

Example 16.1. Assume body temperatures (degrees Fahrenheit) of healthy adults follow a Normal distribution with unknown mean θ and known standard deviation $\sigma = 1$. (It's unrealistic to assume the population standard deviation is known. We'll consider the case of unknown standard deviation later.) Suppose we wish to estimate θ, the population mean healthy human body temperature.
Assume the prior distribution of θ is Normal with mean 98.6 and standard deviation 0.7.

The sample mean body temperature in a sample of 208 healthy adults is 97.7 degrees F.

1. Use the Normal-Normal model to identify the posterior distribution of θ. (Compare to the results of Chapter 8 and Section 14.1.)

2. Find a 98% posterior credible interval.

3. Considering each of the following changes in isolation, indicate if (1) the center and (2) the width of the posterior credible interval would be greater than, less than, or equal to the respective values from the previous part.

   a. Credibility is 80%

b. Sample size is 20

c. Sample mean is 98.0

d. Prior mean is 98.0

e. Prior SD is 0.1

f. Population SD is $\sigma = 0.5$

16.1 Notes

The code below just plugs the numbers into the above Normal-Normal model formulas.

17 Introduction to Multi-Parameter models

- Most interesting problems involve multiple unknown parameters.
- When there are two (or more) unknown parameters the prior and posterior distribution will each be a *joint* probability distribution over *pairs* (or tuples/vectors) of possible values of the parameters.

Example 17.1. Assume body temperatures (degrees Fahrenheit) of healthy adults follow a Normal distribution with unknown mean μ and unknown standard deviation σ. Suppose we wish to estimate both μ, the population mean healthy human body temperature, and σ, the population standard deviation of body temperatures. Assume a discrete prior distribution according to which

- μ takes values 97.6, 98.1, 98.6 with prior probability 0.2, 0.3, 0.5, respectively.
- σ takes values 0.5, 1 with prior probability 0.25, 0.75, respectively.
- μ and σ are independent.

1. Start to construct the Bayes table. What are the possible values of the parameter (in this discrete example)? What are the prior probabilities? (Hint: the parameter θ is a *pair* (μ, σ).)

2. Suppose two temperatures of 97.5 and 97.9 are observed, independently. Identify the likelihood.

3. Complete the Bayes table and find the posterior distribution after observing these two measurements. Compare to the prior distribution.

4. The prior assumes that μ and σ are independent. Are they independent according to the posterior distribution?

Example 17.2. Continuing Example 17.1, let's assume a more reasonable, continuous prior for (μ, σ).

1. What are the possible values of σ? How would you choose a prior for σ?

2. For a prior, assume that: μ has a Normal distribution with mean 98.6 and standard deviation 0.3, σ has a "Half-Normal(0, 1)" distribution (that's Normal(0, 1) but constrained to positive values), and μ and σ are independent. Simulate (μ, σ) pairs from the prior distribution and plot them.

3. Simulate the prior predictive distribution. Do the values seem reasonable? If not, adjust the prior distribution of parameters and try again.

4. Find and interpret a central 98% prior credible interval for μ.

5. Find and interpret a central 98% prior credible interval for σ.

6. What is the prior credibility that both μ and σ lie within their credible intervals?

Example 17.3. Continuing Example 17.1, we'll now use grid approximation to approximate the posterior distribution.

1. Assume a grid of μ values from 96 to 100 in increments of 0.01, and a grid of σ values from 0.01 to 3 in increments of 0.01. How many possible values of the pair (μ, σ) are there; that is, how many rows are there in the Bayes table?

2. Suppose first that we are given a sample of two measurements of 97.9 and 97.5. Use grid approximation to approximate the joint posterior distribution of (μ, σ) Simulate values from the joint posterior distribution and plot them. Compare to the prior. Compute the posterior correlation between μ and σ; are they independent according to the posterior distribution?

3. In a recent study, the sample mean body temperature in a sample of 208 healthy adults was 97.7 degrees F and the sample standard deviation was 1.0 degree F; the sample data is summarized in Section 17.1.5 (it's the same data we used in Section 14.1). Use grid approximation to approximate the joint posterior distribution of (μ, σ) Simulate values from the joint posterior distribution and plot them. Compare to the prior. Compute the posterior correlation between μ and σ; are they independent according to the posterior distribution?

Example 17.4. Continuing Example 17.1, we'll now use simulation to approximate the posterior distribution, given the same sample data as before.

1. Describe how, in principle, you would use a naive (non-MCMC) simulation to approximate the posterior distribution. In practice, what is the problem with such a simulation?

2. Use `brms` to approximate the posterior distribution. Compare to the results of the grid approximation.

3. Find and interpret a central 98% posterior credible interval for μ.

4. Find and interpret a central 98% posterior credible interval for σ.

5. What is the posterior credibility that both μ and σ lie within their credible intervals?

Example 17.5. Continuing the previous example

1. Describe how you could use simulation to approximate the posterior predictive distribution.

2. Run the simulation, summarize the predictive distribution, and find and interpret a 95% posterior predictive interval.

3. Explain how you could use posterior predictive checking to assess the reasonableness of the model in light of the observed data. Does the model seem reasonable?

18 Bayesian Analyis of a Numerical Variable

- When the distribution of the measured variable is symmetric and unimodal, the population mean μ is often the main parameter of interest. However, the population SD σ also plays an important role.
- We have previously assumed that the measured numerical variable follows a conditional Normal(μ, σ) distribution. That is, we assumed a Normal likelihood function.
- However, the assumption of Normality is not always justified, even when the distribution of the measured variable is symmetric and unimodal.

Example 18.1. Assume that birthweights (grams) of human babies follow a Normal distribution with unknown mean μ and unknown standard deviation σ. (1 pound $\approx$ 454 grams.)

1. Considering μ in isolation, what seems like a reasonable prior distribution for μ?

2. What does the parameter σ represent? What is a reasonable prior distribution for σ?

3. Assume a Normal(3400, 100) prior distribution for μ, a Normal(600, 200) prior distribution for σ, and that μ and σ are independent. Explain how you could use simulation to approximate the prior predictive distribution of birthweights. Run the simulation and summarize the results. Does the choice of prior seem reasonable?

Example 18.2. Section 18.1.2 summarizes data on a random sample[1] of 1000 live births in the U.S. in 2001.

1. Inspect the sample data; does it seem reasonable to assume birthweights follow a Normal distribution?

2. Regardless of your answer to the previous question, continue to assume the conditional Normal(μ, σ) model for birthweights, with prior distributions as in the previous example. Use `brms` to find the posterior distribution. Briefly describe the posterior distribution.

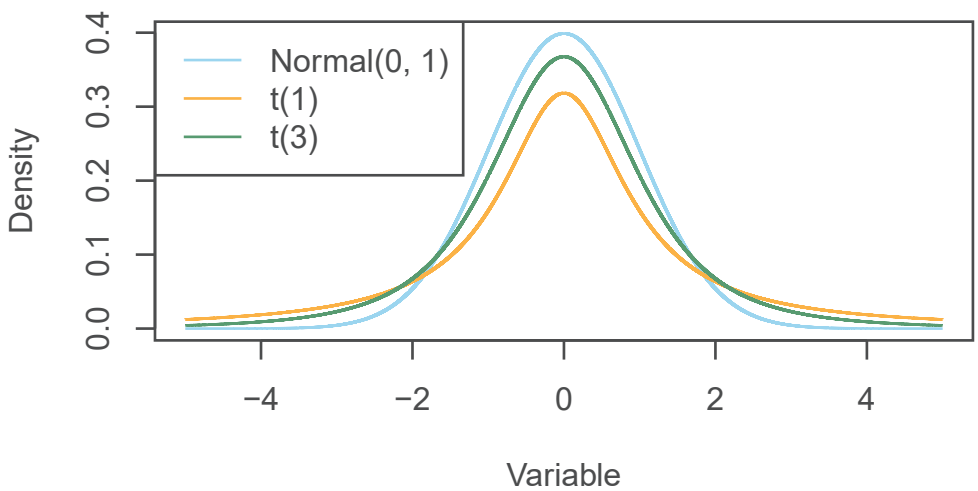
3. How could you use simulation to approximate the posterior predictive distribution of birthweights? Run the simulation and find a 99% posterior prediction interval.

4. What percent of values in the observed *sample* fall outside the prediction interval? What does that tell you?

5. Run a posterior predictive check. Does it indicate any problems?

[1] There are about 4 million live births in the U.S. per year. The data is available at the CDC website. We're only using a random sample to cut down on computation time.

- If the observed data is relatively unimodal and symmetric but has more extreme values than can be accommodated by a Normal likelihood, a t-distribution or other distribution with heavy tails can be used to model the likelihood.
- For "Student" t-distributions, the *degrees of freedom* parameter $\nu \geq 1$ controls how heavy the tails are.
  - When ν is small, the tails are much heavier than for a Normal distribution, leading to a higher frequency of extreme values.
  - As ν increases, the tails get lighter and a t-distribution gets closer to a Normal distribution.
  - For ν greater than 30 or so, there is very little different between a t-distribution and a Normal distribution except in the extreme tails.
- The degrees of freedom parameter ν is sometimes referred to as the "Normality parameter", with larger values of ν indicating a population distribution that is closer to Normal.



Example 18.3. Continuing the birthweight example, we'll now model the distribution of birthweights with a conditional $t(\mu, \sigma, \nu)$ distribution.

1. How many parameters is the likelihood function based on? What are they?

2. What does assigning a prior distribution to the Normality parameter ν represent?

3. Software packages like `brms` will choose a prior for you if you don't specify one. (Basically, the software will run some prior predictive tuning to find a choice of prior that results in a reasonable scale for the response variable.) Change `family` from `gaussian` to `student` and fit the model with `brm`. Run `prior_summary()`; what prior distribution did `brms` assume for `nu`?

4. Consider the posterior distribution for ν. Based on this posterior distribution, is it plausible that birthweights follow a Normal distribution?

5. Consider the posterior distribution for σ. What seems strange about this distribution? (Hint: consider the sample SD.)

6. Using the `brms` output, create a plot of the posterior distribution of $\sigma\sqrt{\frac{\nu}{\nu-2}}$. Does this posterior distribution of the population standard deviation seem more reasonable in light of the sample data?

- The standard deviation of a Normal(μ, σ) distribution is σ.
- However, the standard deviation of a $t(\mu, \sigma, \nu)$ distribution is not σ; rather it is $\sigma\sqrt{\frac{\nu}{\nu-2}} > \sigma$ (as long as $\nu > 2$; if $\nu \leq 2$ then the SD is infinite/undefined).
- When ν is large, $\sqrt{\frac{\nu}{\nu-2}} \approx 1$, and so the standard deviation is approximately σ. However, it can make a difference when ν is small.

Example 18.4. Continuing the previous example.

1. How could you use simulation to approximate the posterior predictive distribution of birthweights? Run the simulation and find a 99% posterior prediction interval. How does it compare to the predictive interval from the model with the Normal likelihood?

2. Run a posterior predictive check. Which model seems to fit the data better: the one based on the Normal likelihood or the t likelihood?

- While we do want a model that fits the data well, we also do not want to risk overfitting the data.
- In particular, we do not want a few extreme outliers to unduly influence the model.
- However, likelihood models based on distributions, like t, with heavier tails than Normal often provide more flexible and robust models.
- Moreover, accommodating the tails often improves the fit in the center of the distribution too.

- In models with multiple parameters, there can be dependencies between parameters, so interpreting the marginal posterior distribution of any single parameter can be difficult.
- It is often more helpful to consider predictive distributions, which accounts for the joint distribution of all parameters.

19 More Bayesian Analyis of a Numerical Variable

Example 19.1. How late do people arrive at parties? FiveThirtyEight conducted a survey to address this question. We'll assume this is a reasonably representative sample and arrival times are measured reliably. Arrival times are measured in minutes after the scheduled start time, rounded to the nearest minute. (Negative arrival times represent arrivals before the scheduled start time.)

1. What would you expect the population distribution of party arrival times to look like? For example, what percent of arrivals would you expect to be early (negative)? Later than two hours? What would you expect of other features like shape, center, and variability?

2. We will start by assuming that, given the values of relevant parameters, arrival times follow a conditional Normal distribution. Describe what this assumption says about arrival times. What are the relevant parameters, and what do they represent? (We will revisit this conditional Normal assumption later, but go with it for now.)

3. Suggest a prior distribution. Explain your reasoning.

4. Describe how you could simulate the prior predictive distribution.

5. This applet conducts the simulation from the previous part. Move the sliders to enter your prior mean and SD for μ and σ. Does the distribution of arrival times seem reasonable based on your expectations? For example, what percent of arrivals are early (negative), or later than two hours — and do these values seem reasonable? If not, revise your assumptions and try again (i.e., play with the sliders in the applet) until you find a distribution that is reasonable. Do not worry about getting it perfect; you just want to settle on assumptions that provide a reasonable starting point. (The assumptions include both the conditional Normal model for arrival times and the prior distribution of parameters. For now we are focusing on the prior distribution, so take the conditional Normal model as given. We'll revisit that assumption later.)

Example 19.2. Section 19.1.2 summarizes the FiveThirtyEight sample[1] of 803 arrival times (rounded to the nearest minute).

We should always first explore the sample data, but to illustrate a point we will not do that yet. Instead, we'll skip straight to fitting the model.

1. Use `brms` to fit the model and summarize the posterior distribution. Briefly describe the posterior distribution.

2. Simulate the posterior predictive distribution. Briefly describe the posterior predictive distribution.

[1] We couldn't find the raw FiveThirtyEight data online, so this data was constructed to match the histogram in the article.

3. Perform a visual posterior predictive check. What does the posterior predictive check say about the appropriateness of our model? Could any problems be fixed by switching from a Normal to a t likelihood?

Example 19.3. The sample data suggests a model that allows for skewness would be more appropriate than Normal. This primarily results in a change of the likelihood function. There are several likelihood functions we can use, but we'll try a Skew Normal family, which introduces an additional skewness parameter α. Since α is a parameter it also needs a prior distribution. Rather than redoing our prior predictive tuning, we'll rely on the fact that software packages like `brms` can utilize default priors.

1. Use `brms` to fit the skew Normal model, letting `brms` choose the prior for α (but specifying the priors for μ and σ as before). What prior does `brms` use?

2. Summarize and describe the posterior distribution.

3. Use simulation to approximate the posterior predictive distribution. Summarize and describe the posterior predictive distribution.

4. Perform a visual posterior predictive check. What does the posterior predictive check say about the appropriateness of the model? Which model seems to be a better fit for the data: likelihood based on Normal or Skew-Normal?

5. Fit the model to the data in `brms` using the `skew_normal` family but letting `brms` choose the prior for all parameters. What prior does `brms` choose? Is the posterior sensitive to the choice of prior?

6. Are the posterior distributions of parameters skewed or relatively symmetric? Is the posterior predictive distribution skewed or relatively symmetric?

7. Write a few sentences summarizing the conclusions of the Bayesian model in context.

19.1 Notes

19.1.1 Prior predictive distribution (Normal likelihood)

```
n_rep = 10000

mu_sim = rnorm(n_rep, 40, sd = 7)
```

20 Comparing Two Means

- Most interesting statistical problems involve multiple measured variables. For example, many problems involve comparing two (or more) populations or groups based on two (or more) samples.
- In such situations, each population or group will have its own parameters, and there will often be dependence between parameters. We are often interested in difference or ratios of parameters between groups.

Example 20.1. Do newborns born to mothers who smoke tend to weigh less at birth than newborns from mothers who don't smoke? We'll investigate this question using birthweight (pounds) data on a sample of full term births in North Carolina over a one year period.

Assume birthweights follow a conditional Normal distribution with mean μ_1 for nonsmokers and mean μ_2 for smokers, and standard deviation σ.

Note: our primary goal will be to compare the means μ_1 and μ_2. We're assuming a common standard deviation σ to simplify a little, but we could (and probably should) also let standard deviation vary by smoking status. (We could also assume something other than conditional Normal, but again, we're simplifying.)

1. Suggest a prior distribution.

2. The prior distribution will be a joint distribution on (μ_1, μ_2, σ) triples. We could assume a prior under which μ_1 and μ_2 are independent. But why might we want to assume some prior *dependence* between μ_1 and μ_2? (For some motivation it might help to consider what the frequentist null hypothesis would be.)

3. Regardless of your answers to the previous parts, assume for the prior: μ_1, μ_2, σ are independent; each of μ_1, μ_2 has a Normal(7.5, 0.5) distribution; σ has a Half-Normal(0, 1) distribution. How do you interpret the parameter $\mu_1 - \mu_2$? Plot the prior distribution of $\mu_1 - \mu_2$, and find prior central 50%, 80%, and 98% credible intervals. Also compute the prior probability that $\mu_1 - \mu_2 > 0$.

4. Briefly describe the sample data, summarized in Section 20.1.2.

5. Is it reasonable to assume that the two *samples* are independent? (In this case the y_1 and y_2 samples would be *conditionally independent* given (μ_1, μ_2, σ).)

6. Describe how you would compute the likelihood. For concreteness, how you would you compute the likelihood if there were only 4 babies in the sample: 2 non-smokers with birthweights of 8 pounds and 7 pounds, and 2 smokers with birthweights of 8.3 pounds and 7.1 pounds.

7. Use `brms` to approximate the posterior distribution. Are μ_1 and μ_2 independent according the posterior? Why do you think that is?

8. Plot the posterior distribution of $\mu_1 - \mu_2$ and describe it. Compute and interpret posterior central 50%, 80%, and 98% credible intervals. Also compute and interpret the posterior probability that $\mu_1 - \mu_2 > 0$.

9. If we're interested in $\mu_1 - \mu_2$, why didn't we put a prior directly on $\mu_1 - \mu_2$ rather than on (μ_1, μ_2)?

10. Plot the posterior distribution of μ_1/μ_2, describe it, and find and interpret posterior central 50%, 80%, and 98% credible intervals.

11. Is there some evidence that babies whose mothers smoke tend to weigh less than those whose mothers don't smoke?

12. Can we say that smoking is the cause of the difference in mean weights?

13. Is there some evidence that babies whose mothers smoke tend to weigh *much* less than those whose mothers don't smoke? Explain.

14. One quantity of interest is the **effect size**, which is a way of measuring the magnitude of the difference between groups. When comparing two means, a simple measure of effect size (*Cohen's d*) is

$$\frac{\mu_1 - \mu_2}{\sigma}$$

Plot the posterior distribution of this effect size and describe it. Compute and interpret posterior central 50%, 80%, and 98% credible intervals.

Independence in the data versus in the prior/posterior

It is typical to assume *independence in the data*, e.g., independence of values of the measured variables within and between samples (conditional on the parameters). Whether independence in the data is a reasonable assumption depends on how the data is collected.

But whether it is reasonable to assume prior *independence of parameters* is a completely separate question and is dependent upon our subjective beliefs about any relationships between parameters.

Transformations of parameters

The primary output of a Bayesian data analysis is the full joint posterior distribution on all parameters. Given the joint distribution, the distribution of transformations of the primary parameters is readily obtained.

Effect size for comparing means

When comparing groups, a more important question than "is there a difference?" is "*how large* is the difference?" An **effect size** is a measure of the magnitude of a difference between groups. A difference in parameters can be used to measure the absolute size of the difference in the measurement units of the variable, but effect size can also be measured as a relative difference.

When comparing a numerical variable between two groups, one measure of the population effect size is **Cohens's** d

$$\frac{\mu_1 - \mu_2}{\sigma}$$

The values of any numerical variable vary naturally from unit to unit. The SD of the numerical variable measures the degree to which individual values of the variable vary naturally, so the

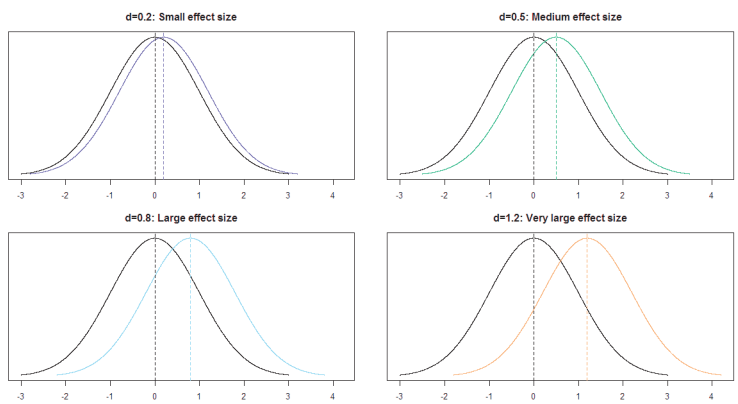
SD provides a natural "scale" for the variable. Cohen's d compares the magnitude of the difference in means relative to the natural scale (SD) for the variable.

Some rough guidelines for interpreting $|d|$:

d	0.2	0.5	0.8	1.2	2.0
Effect size	Small	Medium	Large	Very Large	Huge

For example, assume the two population distributions are Normal and the two population standard deviations are equal. Then when the effect size is 1.0 the median of the distribution with the higher mean is the 84th percentile of the distribution with the lower mean, which is a very large difference.

d	0.2	0.5	0.8	1.0	1.2	2.0
Effect size	Small	Medium	Large		Very Large	Huge
Median of population 1 is (blank) percentile of population 2	58th	69th	79th	84th	89th	98th



20.1 Notes

20.1.1 Prior distribution of $\mu_1 - \mu_2$

The prior distribution of $\mu_1 - \mu_2$ is Normal with mean 7.5-7.5 = 0 and standard deviation $\sqrt{0.5^2 + 0.5^2} = 0.707$

```
qnorm(c(0.01, 0.10, 0.25, 0.75, 0.90, 0.99),
      mean = 7.5 - 7.5,
```

21 Introduction to Markov Chain Monte Carlo (MCMC) Simulation

Markov chain Monte Carlo (MCMC) methods provide powerful and widely applicable algorithms for simulating from probability distributions, including complex and high-dimensional posterior distributions.

Example 21.1. A politician campaigns on a long east-west chain of islands[1]. At the end of each day she decides to stay on her current island, move to the island to the east, or move to the island to the west. Her goal is to plan her daily island visits proportional to the islands' populations, so that she spends the most days on the most populated island, and proportionally fewer days on less populated islands. Unfortunately, (1) she doesn't know how many islands there are, and (2) she doesn't know the population of each island. However, when she visits an island she can determine its population. And she can send a scout to the east/west adjacent islands to determine their population before visiting. How can the politician achieve her goal of visiting islands proportionally to their population in the long run?

Suppose that every day, the politician makes her travel plans according to the following algorithm.

- She flips a fair coin to *propose* travel to the island to the east (heads) or west (tails). (If there is no island to the east/west, treat it as an island with population zero below.)
- If the proposed island has a population greater than that of the current island, then she travels to the proposed island.
- If the proposed island has a population less than that of the current island, then:
  - She computes a, the ratio of the population of the proposed island to the current island.
  - She travels to the proposed island with probability a,
  - And with probability $1 - a$ she spends another day on the current island.

Suppose that, unbeknownst to the politician, are 5 islands, labeled $1, \ldots, 5$ from west to east, and that island θ has population θ (thousand). (Remember, the politician would not know this information.)

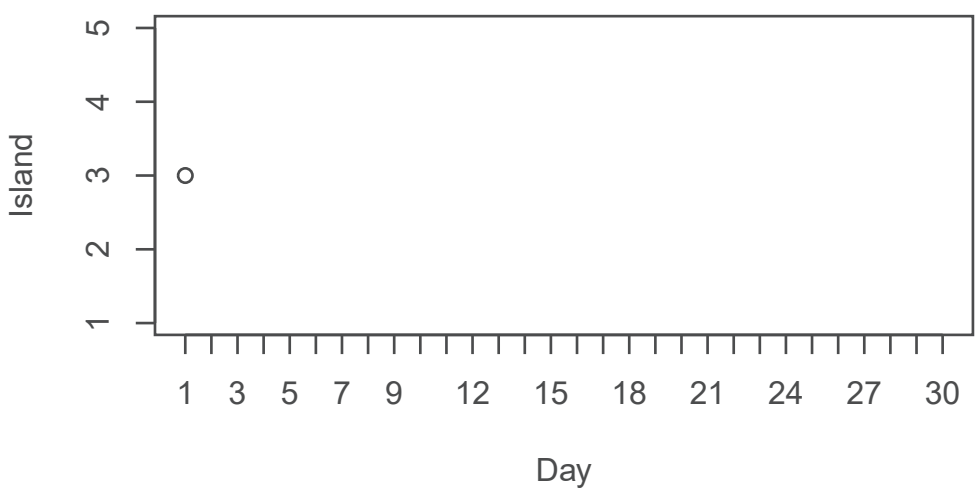
[1]This island hopping example is inspired by Kruschke, *Doing Bayesian Data Analysis*.

Suppose she starts on island $\theta = 3$ on day 1.

(11) What are her possible locations on day 2? What are the relative populations of these locations?

(12) For each of the possible day 2 locations, what is the probability that she visits the location on day 2?

(13) How could you use a coin and a spinner to simulate *by hand* the politician's movements over a number of days? Conduct the simulation and plot the path of her movements.



(14) Construct *by hand* a plot displaying the simulated relative frequencies of days in each of the 5 islands.

(15) Now use code to simulate the politician's movements for many days. Plot the island path.

(16) Plot the simulated relative frequencies of days in each of the 5 islands.

(17) Recall the politician's goal of visiting each island in proportion to its population. Given her goal, what should the plot from the previous part look like? Does the algorithm result in a reasonable approximation?

(18) Is the next island visited dependent on the current island?

(19) Is the next island visited dependent on *how she got to* the current island? That is, given the current island is her next island independent of her past history of previous movements?

- The goal of a Markov chain Monte Carlo method is to simulate from a probability distribution of interest. In Bayesian contexts, the distribution of interest will usually be the posterior distribution of parameters given data.
- A **Markov chain** is a random process that exhibits a special "one-step" dependence structure. Namely, conditional on the most recent value, any future value is conditionally independent of any past values.
- The idea of MCMC is to build a Markov chain whose long run distribution — that is, the distribution of state visits after a large number of "steps" — is the probability distribution of interest.
- Then we can *indirectly* simulate a representative sample from the probability distribution of interest, and use the simulated values to approximate the distribution and its characteristics, by running an appropriate Markov chain for a sufficiently large number of steps.
- The Markov chain does not need to be fully specified in advance, and is often constructed "as you go" via an algorithm like a "modified random walk".
- Each step of the Markov chain typically involves
  - A **proposal** for the next state, which is generated according to some known probability distribution or mechanism,
  - A **decision of whether or to accept the proposal**. The decision usually involves probability
    - With probability a, accept the proposal and step to the next state
    - With probability $1-a$, reject the proposal and remain in the current state for the next step.

Example 21.2. Continuing Example 21.1. Suppose again that the number of islands and their populations is unknown, but that the population for island θ is proportional to $\theta^2 e^{-0.5\theta}$, where the islands are labeled from west to east 1, 2, ... That is, when she or the scout visits island θ they only determine that the population is $c\theta^2 e^{-0.5\theta}$ for some value of c, but they do not learn what c is For example, when they visit island $\theta = 1$ they learn its population is $c(1^2)e^{-0.5(1)} \approx 0.606c$, when they visit island $\theta = 2$ they learn its population is $c(2^2)e^{-0.5(2)} \approx 1.472c$, etc, but they never learn what c is.

They'll attempt the same algorithm as in Example 21.1. Suppose she starts on island $\theta = 3$ on day 1.

1. What are her possible locations on day 2? What are the relative populations of these locations?

2. For each of the possible day 2 locations, what is the probability that she visits the location on day 2?

3. Can she implement her algorithm in this case? That is, is it sufficient to learn that the populations are *in proportion to* $\theta^2 e^{-0.5\theta}$ without knowing the actual populations? In particular, is there enough information to compute the "acceptance probability" a?

4. Now use code to simulate the politician's movements for many days. Plot the island path.

5. Plot the simulated relative frequencies of days in each of the islands.

6. Recall the politician's goal of visiting each island in proportion to its population. Given her goal, what should the plot from the previous part look like? Does the algorithm result in a reasonable approximation?

7. Why doesn't the politician just always travel to or stay on the island with the larger population?

8. What would happen if there were an island among the east-west chain with population 0? (For example, suppose there are 10 islands, she starts on island 1, and island 5 has population 0.) How could she modify her algorithm to address this issue?

- The goal of an MCMC method is to simulate θ values from a probability distribution $\pi(\theta)$.
- One commonly used MCMC method is the **Metropolis algorithm**[2]
- To generate θ_{new} given θ_{current}:

1. Given the current value θ_{current} propose a new value θ_{proposed} according to the *proposal (or "jumping") distribution j.*
$$j(\theta_{\text{current}} \to \theta_{\text{proposed}})$$
is the conditional density that θ_{proposed} is proposed as the next state given that θ_{current} is the current state
2. Compute the *acceptance probability* as the ratio of target density at the current and proposed states
$$a(\theta_{\text{current}} \to \theta_{\text{proposed}}) = \min\left(1,\ \frac{\pi(\theta_{\text{proposed}})}{\pi(\theta_{\text{current}})}\right)$$
3. Accept the proposal with probability $a(\theta_{\text{current}} \to \theta_{\text{proposed}})$ and set $\theta_{\text{new}} = \theta_{\text{proposed}}$. With probability $1 - a(\theta_{\text{current}} \to \theta_{\text{proposed}})$ reject the proposal and set $\theta_{\text{new}} = \theta_{\text{current}}$.
   - If $\pi(\theta_{\text{proposed}}) \geq \pi(\theta_{\text{current}})$ then the proposal will be accepted with probability 1.
   - Otherwise, there is a positive probability or rejecting the proposal and remaining in the current state. But this still counts as a "step" of the MC.

- The Metropolis-Hastings algorithm works for any proposal distribution which allows for eventual access to all possible values of θ. That is, if we run the algorithm long enough

[2]The algorithm is named after Nicholas Metropolis, a physicist who led the research group which first proposed the method in the early 1950s (though Metropolis himself may have had little to do with the actual invention of the algorithm).

then the distribution of the simulated values of θ will approximate the target distribution $\pi(\theta)$. Thus we can choose a proposal distribution that is easy to simulate from.

- However, in practice the choice of proposal distribution is extremely important — especially when simulating from high dimensional distributions — because it determines *how fast* the MC converges to its long run distribution.
- There are a wide variety of MCMC methods and their extensions that strive for computational efficiency by making "smarter" proposals. These algorithms include Gibbs sampling, Hamiltonian Monte Carlo (HMC), and No U-Turn Sampling (NUTS). We won't go into the details of the many different methods available. Regardless of the details, all MCMC methods are based on the two core principles of proposal and acceptance.

Example 21.3. Suppose we wish to estimate θ, the proportion of Cal Poly students who have read a non-school related book in 2023. Assume a Beta(1, 3) prior distribution for θ. In a sample of 25 Cal Poly students, 4 have read a book in 2023. We'll use MCMC to approximate the posterior distribution of θ.

(Note: this is a situation where we can work out the posterior analytically, so we don't need MCMC. But this example allows us to see how MCMC is implemented in a relatively simple situation where we can check if the algorithm works.)

1. Without actually computing the posterior distribution, what can we say about it based on the assumptions of the model?

2. What are the possible "states" that we want our Markov chain to visit?

3. Given a current state θ_{current} how could we *propose* a new value θ_{proposed}, using a continuous analog of "random walk to neighboring states"?

4. How would we decide whether to accept the proposed move? How would we compute the probability of accepting the proposed move?

5. Suppose the current state is $\theta_{\text{current}} = 0.20$ and the proposed state is $\theta_{\text{proposed}} = 0.15$. Compute the probability of accepting this proposal.

6. Suppose the current state is $\theta_{\text{current}} = 0.20$ and the proposed state is $\theta_{\text{proposed}} = 0.25$. Compute the probability of accepting this proposal.

7. Use code to run the algorithm and simulate many values of θ. Plot the simulated values of θ as a function of step (called a "trace plot"), and summarize the simulated values of θ in a histogram.

8. What is the theoretical posterior distribution of θ? Does the distribution of values of θ simulated by the MCMC algorithm provide a reasonable approximation to the theoretical posterior distribution?

- We will most often use MCMC methods to simulate values from a posterior distribution $\pi(\theta|y)$ of parameters θ given data y.

- The Metropolis (or Metropolis-Hastings algorithm) allows us to simulate from a posterior distribution *without computing the posterior distribution.*
- Recall that the inputs of a Bayesian model are: (1) the data y; (2) the model for the data, $f(y|\theta)$, which determines the likelihood; and (3) the prior distribution $\pi(\theta)$ of parameters θ.
- The target posterior distribution satisfies

$$\pi(\theta|y) \propto f(y|\theta)\pi(\theta)$$

- Therefore, the acceptance probability of the Metropolis algorithm can be computed based on the form of the prior and likelihood alone

$$a(\theta_{\text{current}} \to \theta_{\text{proposed}}) = \min\left(1, \frac{\pi(\theta_{\text{proposed}}|y)}{\pi(\theta_{\text{current}}|y)}\right) = \min\left(1, \frac{f(y|\theta_{\text{proposed}})\pi(\theta_{\text{proposed}})}{f(y|\theta_{\text{current}})\pi(\theta_{\text{current}})}\right)$$

- Furthermore, the product of prior and likelihood only needs to be evaluated for the states the chain actual visits. This is especially helpful in high dimensional models with many parameters. (Recall that when there are many parameters, grid approximation becomes infeasible simply because there are too many rows in the grid to actually compute the product of prior likelihood for.)
- As the chain progresses, the algorithm fills in more and more of the relative posterior plausibilities; e.g., the current state is blank times more/less plausible than each of the states that preceded it.

Example 21.4. We have seen how to estimate a process probability p in a Binomial situation, but what if the number of trials is also random? For example, suppose we want to estimate both the average number of three point shots Steph Curry attempts per game (μ) and the probability of success on a single attempt (p). Assume that

- Conditional on μ, the number of attempts in a game N has a Poisson(μ) distribution
- Conditional on n and p, the number of successful attempts in a game Y has a Binomial(n, p) distribution.
- Conditional on (μ, p), the values of (N, Y) are independent from game to game

For the prior distribution, assume

- μ has a Gamma(10, 2) distribution
- p has a Beta(4, 6) distribution
- μ and p are independent

In two recent games, Steph Curry made 4 out of 10 and 6 out of 11 attempts.

1. What is the likelihood function?

2. Without actually computing the posterior distribution, what can we say about it based on the assumptions of the model?

3. What are the possible "states" that we want our Markov chain to visit?

4. Given a current state θ_{current} how could we *propose* a new value θ_{proposed}, using a continuous analog of "random walk to neighboring states"?

5. Suppose the current state is $\theta_{\text{current}} = (8, 0.5)$ and the proposed state is $\theta_{\text{proposed}} = (7.5, 0.55)$. Compute the probability of accepting this proposal.

6. Use code to run the algorithm. Plot the simulated values of θ in a trace plot, and summarize them in approximate plots.

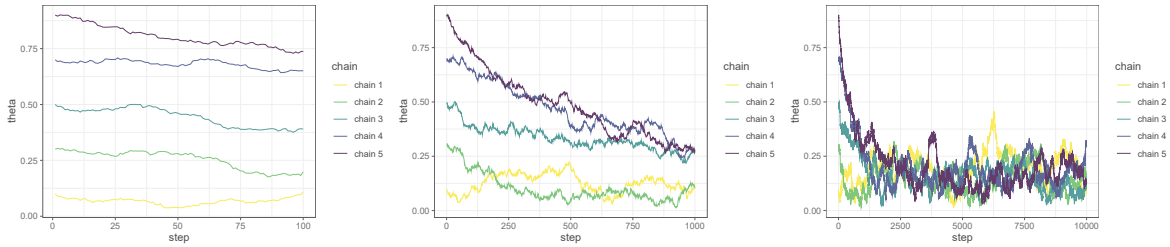
22 Some Diagnostics for MCMC Simulation

- The goal we wish to achieve with MCMC is to simulate from a probability distribution of interest (e.g., the posterior distribution).
- The idea of MCMC is to algorithmically build a Markov chain whose long run distribution is the probability distribution of interest. Then we can simulate a sample from the probability distribution, and use the simulated values to summarize and investigate characteristics of the probability distribution, by running the Markov chain for a sufficiently large number of steps.
- But how do we know that the MCMC algorithm works? Some questions of interest:
  - Does the algorithm produce samples that are *representative* of the target distribution of interest?
  - Are estimates of characteristics of the distribution (e.g., posterior mean, posterior standard deviation, 98% credible region) based on the simulated Markov chain *accurate* and stable?
  - Is the algorithm *efficient*, in terms of time or computing power required to run?

Example 22.1. Recall Example 21.3 in which we used a Metropolis algorithm to simulate from a Beta(5, 24) distribution. Given current state θ_c, the proposal was generated from a $N(\theta_c, \delta)$ distribution, where δ was a specified value (determining what constitutes the "neighborhood" in the continuous random walk analog). The algorithm also needs some initial value of θ to start with; in Example 21.3 we used an initial value of 0.5. What is the impact of the initial value?

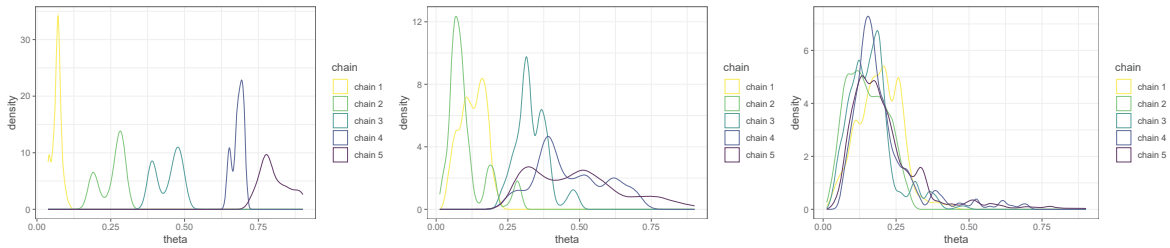
The following plots display the values of the first 100 steps, 1000 steps, and 10000 steps — and their densities — for 5 different runs of the Metropolis chain each starting from a different initial value: 0.1, 0.3, 0.5, 0.7, 0.9. The value of δ is 0.005. (Note: we're setting the δ value to be very small to illustrate a point.) What do you notice in the plots? Does the initial value influence the results?

step	chain 1	chain 2	chain 3	chain 4	chain 5
1	0.1000000	0.3000000	0.5000000	0.7000000	0.9000000
2	0.0894830	0.3036224	0.4909961	0.6918153	0.8951483
3	0.0888948	0.3002659	0.4919129	0.6946801	0.9008528
4	0.0853845	0.3030381	0.4871295	0.6898734	0.8989818
5	0.0826313	0.2949438	0.4761009	0.6872206	0.8989818
6	0.0768894	0.2973711	0.4788800	0.6872206	0.8996724
7	0.0796352	0.2922830	0.4724440	0.6858423	0.8935327
8	0.0807874	0.2964747	0.4742603	0.6898477	0.8863538
9	0.0790930	0.2964846	0.4709396	0.6887288	0.8897852
10	0.0783251	0.2891121	0.4727961	0.6856081	0.8851058



(a) First 100 steps for each chain (b) First 1000 steps for each chain (c) First 10000 steps for each chain

Figure 22.1: Trace plots for five chains with different starting states



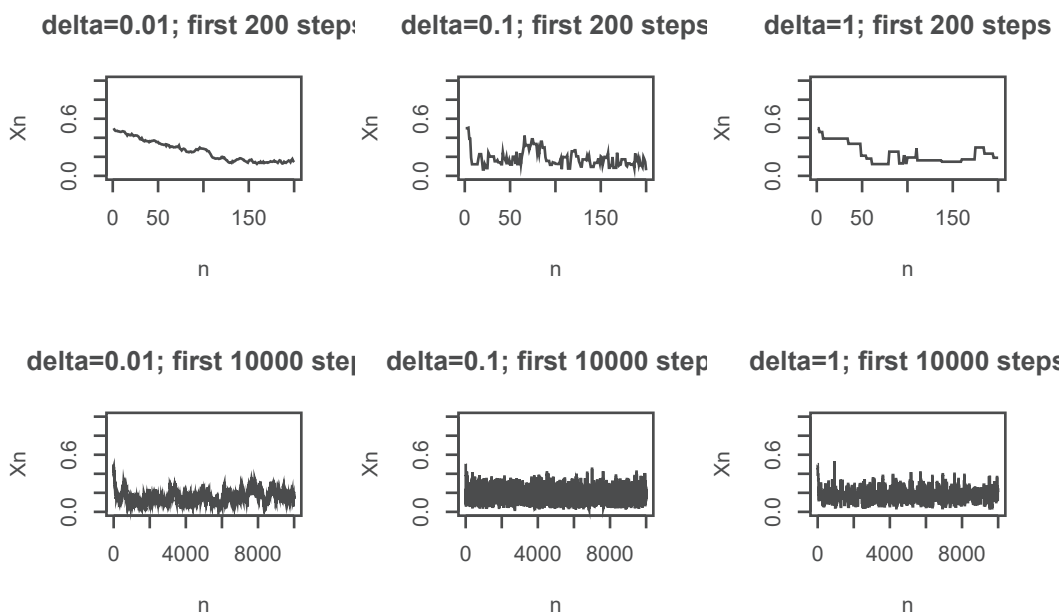
(a) First 100 steps for each chain (b) First 1000 steps for each chain (c) First 10000 steps for each chain

Figure 22.2: Density plots for five chains with different starting states

- The goal of an MCMC simulation is to simulate a *representative* sample of values from the target distribution.
- While an MCMC algorithm should converge eventually to the target distribution, it might take some time to get there.
- **Warm up** (or "warm up") refers to the process of discarding the first several hundred or thousand steps of the chain to allow for a "settling in" period. Only values simulated after the warm up period are used to approximate the target distribution.
- In practice, it is common to use a warm up period of several hundred or thousands of steps.
- To get a better idea of how long the warm up period should be, run the chain starting from several disperse initial conditions to see how long it takes for the paths to "overlap".
- After warm up period, examine trace plots or density plots for multiple chains; if the plots do not "overlap" then there is evidence that the chains have not converged, so they might not be producing representative samples from the target distribution, and therefore a longer warm up period is needed.
- In `brms`:
  - `iter` is the total number of steps for each single chain
  - `warmup` is the number of steps to discard
  - `chains` is the number of chains to run
  - For example `iter = 3500`, `warmup = 1000`, `chains = 4` will produce 3500 - 1000 = 2500 values for each of 4 chains, resulting in 10000 simulated values in total
- The **Gelman-Rubin statistic** (a.k.a., shrink factor, "R-hat") is a numerical check of convergence based on a measure of variance between chains relative to variability with chains.
- The idea is that if multiple chains have settled into representative sampling, then the average difference between chains should be equal to the average difference within chains (i.e., across steps). Roughly, if multiple chains are all producing representative samples from the target distribution, then given a current value, it shouldn't matter if you take the next value from the chain or if you hop to another chain.
- Thus, after the warm up period, the shrink factor should be close to 1.

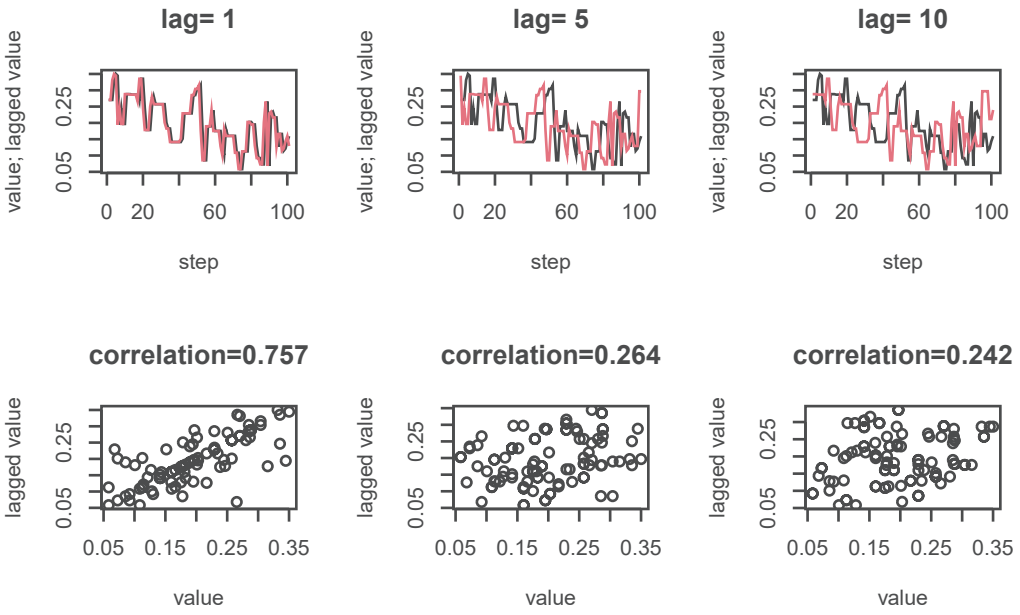
Example 22.2. Continuing Example 22.1. Recall that we used a Metropolis algorithm to simulate from a Beta(5, 24) distribution. Given current state θ_c, the proposal was generated from a $N(\theta_c, \delta)$ distribution, where δ was a specified value (determining what constitutes the "neighborhood" in the continuous random walk analog).

The following plots display the results of three different runs, one each for $\delta = 0.01$, $\delta = 0.1$, $\delta = 1$, all with an initial value of 0.5. Describe the differences in the behavior of the chains. Which chain seems "best"? Why?

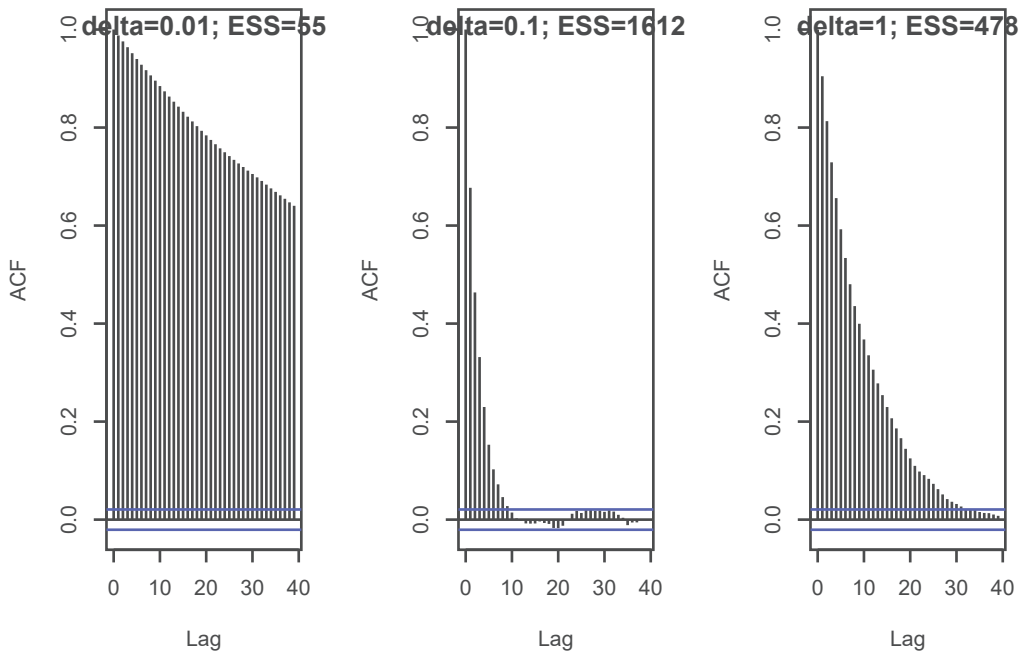


- The values of a Markov chain at different steps are *dependent*.
- If the degree of dependence is too high, the chain will tend to get "stuck", requiring a large number of steps to fully explore the target distribution of interest.

Example 22.3. Continuing Example 22.2 with $\delta = 0.1$, the following plots display the actual values of the chain (after warm up) and the values lagged by 1, 5, and 10 time steps. Are the values at different steps dependent? In what way are they not too dependent?



- An autocorrelation plot displays the autocorrelation within in a chain as a function of lag.
- If the ACF takes too long to decay to 0, the chain exhibits a high degree of dependence and will tend to get stuck in place.



Example 22.4. Continuing Example 22.2 involving Metropolis sampling from a Beta(5, 24) posterior distribution. We know that the posterior mean is $5/29 = 0.172$. But what if we want to approximate this via simulation?

1. Suppose you simulated 10000 *independent* values from a Beta(5, 24) distribution, e.g. using `rbeta`. How would you use the simulated values to estimate the posterior mean?

2. What is the *standard error* of your estimate from the previous part? What does the standard error measure? How could you use simulation to approximate the standard error?

3. Now suppose you simulated 10000 from a Metropolis chain (after warm up). How would you use the simulated values to estimate the posterior mean? What does the standard error measure in this case? Could you use the formula from the previous part to compute the standard error? Why?

4. Consider the three chains in Example Example 22.2 corresponding to the three δ values 0.01, 0.1, and 1. Which chain provides the most reliable estimate of the posterior mean? Which chain yields the smallest standard error of this estimate?

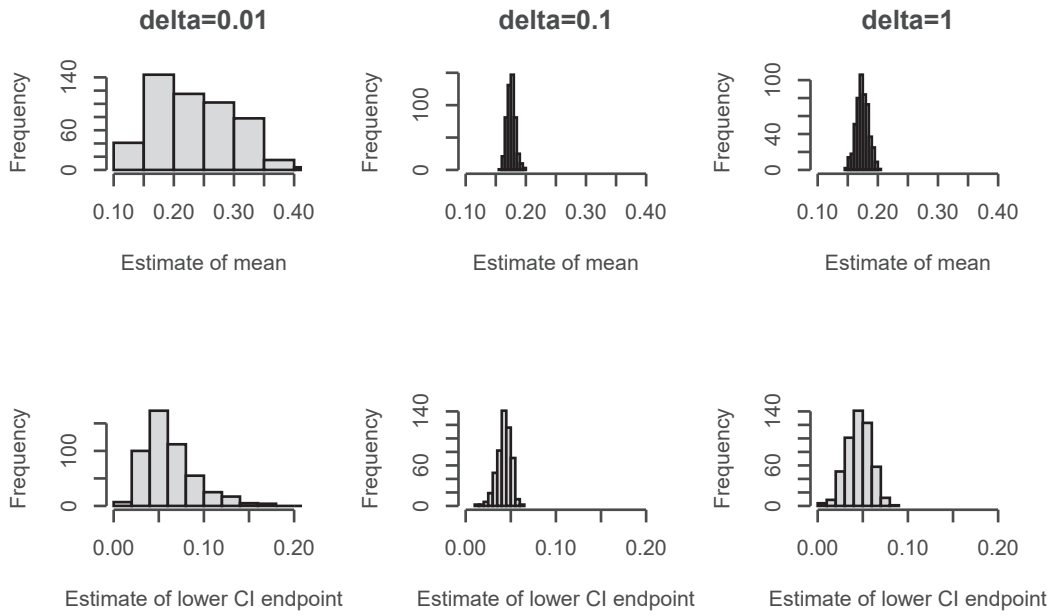
- The **effective sample size (ESS)** is a measure of how much independent information there is in an autocorrelated chain.
- The effective sample size of a chain with N steps (after warm up) is

$$\text{ESS} = \frac{N}{1 + 2\sum_{\ell=1}^{\infty} \text{ACF}(\ell)}$$

  where the infinite sum is typically cut off at some upper lag (say $\ell = 20$).

  - For a completely independent chain, the autocorrelation would be 0 for all lags and the ESS would just be the number of steps N.
  - The more quickly the ACF decays to 0, the larger the ESS.
  - The more slowly the ACF decays to 0, the smaller the ESS.

- The **Monte Carlo standard error (MCSE)** is the standard error of a statistic generated based on an MCMC algorithm.
- The MCSE of a statistic measures the run-to-run variability of values of the statistic over many runs of the chain of the same number of steps.
- For many statistics (means, proportions) the MCSE based on a chain with effective sample size ESS is on the order of $\frac{1}{\sqrt{\text{ESS}}}$.



22.1 Notes

This is the `brms` code for the Beta-Binomial example in this section, including some of the diagnostic plots and statistics.

```
library(brms)
```

```
Loading required package: Rcpp

Loading 'brms' package (version 2.18.0). Useful instructions
can be found by typing help('brms'). A more detailed introduction
to the package is available through vignette('brms_overview').


Attaching package: 'brms'

The following object is masked from 'package:stats':

    ar
```

23 Bayesian Simple Linear Regression Model

Regression is one of the most widely used statistical techniques for modeling relationships between variables. We will now consider a Bayesian treatment of *simple linear regression*. We'll use the following example throughout.

Percent body fat (PBF, total mass of fat divided by total body mass) is an indicator of physical fitness level, but it is difficult to measure accurately. Body mass index (BMI), which is easily computed based on height and weight ($\text{BMI} = \frac{\text{weight in kg}}{(\text{height in m})^2}$), is another commonly used measure[1]. What is the relationship between BMI and percent body fat? How can we use a person's BMI to estimate their percent body fat?

We'll fit a Bayesian model for predicting percent body fat from BMI using data on a nationally representative sample of 2157 U.S. males from the National Health and Nutrition Examination Survey (NHANES).

Example 23.1. Consider the assumptions of a Bayesian simple linear regression model for predicting percent body fat from BMI.

1. What are the assumptions of the simple linear regression model? (There are many different regression models. We're referring to the one that is usually the first one you see in introductory statistics. Sometimes the assumptions are summarized with the acronym "LINE".)

2. What are the parameters in the model? That is, what do we need to put a prior distribution on (in order to eventually get a posterior distribution given sample data)? What do the parameters represent in context?

[1] But BMI has its own problems; see http://fivethirtyeight.com/features/bmi-is-a-terrible-measure-of-health/.

3. Suggest a prior distribution. Explain your reasoning.

4. Describe in full detail how you would simulate the prior predictive distribution of percent body fat for males with a BMI of 20.

5. Simulate the prior predictive distribution of percent body fat for a range of BMI values. Do the distributions seem reasonable? If not, experiment with different priors until you find one that produces a reasonable prior predictive distribution.

- Consider data on a numerical response variable y and a numerical explanatory (or predictor) variable x. A **simple linear regression model** of y on x assumes:
  - *Linearity*: the conditional mean of y is *linear* in x; for some slope β_1 and intercept β_0:
  $$\mathrm{E}(y|x) = \beta_0 + \beta_1 x$$
  - *Independence*: $y_1, \ldots, y_n$ are independent[2]
  - *Normality*: for each x, the conditional distribution of y given x is Normal
  - *Equal variance*: the conditional variance of y does not depend on x
  $$\mathrm{Var}(y|x) = \sigma^2$$
- The regression model is *conditional*. That is, we make assumptions about and perform inference for the conditional distribution of the response variable y given values of the explanatory variable x. Therefore, even when we technically have a sample of pairs

[2] In a Bayesian context, we assume conditional independence of $y_1, \ldots, y_n$ given the parameters β_0, β_1, σ

(x_i, y_i) from a bivariate distribution, we will always condition on the values of x and treat them as non-random.

- Linearity is an assumption: we assume there is some "true" population *regression line* $\beta_0 + \beta_1 x$ which specifies the conditional *mean* of the response variable for each value of the explanatory variable.
- The above set of assumptions can be stated more compactly as

$$y_i \sim N\left(\beta_0 + \beta_1 x_i, \sigma^2\right), \text{ and } y_1, \ldots, y_n \text{ are independent}$$

Equivalently,

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \text{ where } \varepsilon_i, \ldots, \varepsilon_n \text{ are i.i.d. } N(0, \sigma).$$

- The parameters of the above model are β_0, β_1, and σ.
- σ_y^2 denotes the variance of the response variable y with respect to its overall mean, while σ^2 denotes the variance of values of the response variable about the true regression line. Thus, $\sigma^2 = (1 - \rho^2)\sigma_y^2$, where ρ is the population correlation between the explanatory and response variables.

Example 23.2. Section 23.1.3 summarizes the sample data. Before fitting a Bayesian model, we'll examine the usual frequentist results.

1. Find and interpret the slope of the fitted regression line.

2. Find and interpret a 95% confidence interval for the slope of the regression line. (How do you interpret 95% confidence?)

3. Find the equation of the fitted regression line.

4. Find and interpret the fitted value for men with a BMI of 20 kg/m^2.

5. Find and interpret a 95% confidence interval based on the fitted value for men with a BMI of 20 kg/m^2.

6. Find and interpret a point estimate of σ.

7. Find and interpret a 95% *prediction* interval for men with a BMI of 20 kg/m^2.

Example 23.3. Now we'll fit a Bayesian model and find the posterior distribution.

1. Suppose there are just two observations of (BMI, PBF): (15, 20) and (40, 30). How would you compute the likelihood?

2. Use `brms` to fit the model to the full sample data. Run some diagnostic checks; does the MCMC algorithm seem to be working efficiently?

3. Inspect the joint posterior distribution of β_0 and β_1. What do you notice? Why do you think this happens?

4. Now suppose we replace BMI (x) with its centered version ($x - \bar{x}$). What does the parameter β_0 represent in this model? Suggest a prior distribution for β_0 (considering it in isolation).

5. Use `brms` to fit the model using centered BMI as the explanatory variable. Inspect the joint posterior distribution of β_0 and β_1. What do you notice? Why do you think this happens?

Example 23.4. Now we'll summarize and report conclusions based on the posterior distribution, using the model with centered BMI.

1. Describe the posterior distribution of β_1. Find and interpret 50%, 80%, and 98% posterior credible intervals for β_1.

2. What does the parameter $\mu_{20} = \beta_0 + \beta_1(20 - \bar{x})$ represent in context? Explain how you could approximate the posterior distribution of μ_{20}. Then approximate the posterior distribution, describe it, and find and interpret a 98% posterior credible interval for μ_{20}.

3. Describe the posterior distribution of σ. Find and interpret 50%, 80%, and 98% posterior credible intervals for σ.

4. Compare the results of the Bayesian and frequentist analyses. How are they similar? How are they different?

Example 23.5. Now we'll use the fitted model to predict percent body fat.

1. Explain how you could approximate the posterior predictive distribution of percent body fat for men with a BMI of 20 kg/m^2.

2. Find and interpret a 95% posterior prediction interval for men with a BMI of 20 kg/m^2.

3. Compare the 95% posterior prediction interval with the corresponding frequentist 95% prediction interval. How are they similar? How are they different?

4. Suggest how you could perform a posterior predictive check. Then use software to perform the check. Does the model seem reasonable based on the data?

- There are a number of ways the assumptions of the simple linear regression model could be revised.
- We could assume the mean of y changes with x is a non-linear way (e.g., quadratic).
- We could assume distributions other than Normal in the likelihood. For example, we could assume a t-distribution if outliers were an issue.
- We could assume the conditional variance of y also changes with x.
- We could include multiple explanatory variables x.
- Each of the above would be a change in the *likelihood*, perhaps introducing additional parameters.

Example 23.6. Now we'll investigate sensitivity of the results to choice of prior. Rerun the model (using centered BMI) but let `brms` choose the prior. Compare the results of the two Bayesian models; are the results very sensitive to the choice of prior?

24 Introduction to Bayesian Hierarchical Models

In a **Bayesian hierarchical**[1] **model**, the model for the data depends on certain parameters, and those parameters in turn depend on other parameters. Hierarchical modeling provides a framework for building complex and high-dimensional models from simple and low-dimensional building blocks.

Suppose we want to estimate the true, long run free throw percentage for each basketball player currently on the Philadelphia 76ers[2]. That is, we want to estimate each player's true probability of successfully scoring on any free throw attempt. We'll consider data from games in the 2022-2023 NBA season obtained from basketball-reference as of Feb 27, summarized in Section 24.1.1.

Example 24.1. Let θ_j denote player j's true (long run) FT percentage; for example, θ_{Embiid}, θ_{Harden}, etc. Let n_j denote the number of free throw attempts and y_j the number of free throws made for player j.

1. One way to estimate θ_j is to pool all the data together. Over the 19 players in the data set, there are 1482 attempts, of which 1229 were successful, for an overall free throw percentage of 82.9%. The sample mean free throw percentage, among players with nonzero attempts, is 77.7%. So we could just estimate each θ_j with the overall free throw percentage (or the sample mean free throw percentage). What are some issues with this approach?

2. At the other extreme, we could estimate θ_j using just the attempts for player j. For example, use only the attempts for Embiid to estimate θ_{Embiid}, only the attempts for Harden to estimate θ_{Harden}, etc. What are some issues with this approach? (Hint:

[1] Hierarchical models are often called *multilevel, mixed effects, or random effects models.*

[2] Along with one bonus "player" who will illustrate some ideas.

consider the players with no attempts, or few attempts.)

3. In a Bayesian model, the θ_j's would have a prior distribution. We could assume a Bayesian model with a prior in which all θ_j are independent. What are some issues with this approach?

Example 24.2. Let's first consider just a single player with true (long run) free throw percentage θ. Given n independent attempts, conditional on θ, we might assume that y follows a Binomial(n, θ) distribution.

1. If we imagine the player is randomly selected from among current NBA players, how might we interpret the prior distribution for θ?

2. The prior distribution for θ has a prior mean; let's denote it μ. How might we interpret μ?

3. The prior mean μ is also a parameter, so we could place a prior distribution on it. What would the prior distribution on μ represent?

4. In this model μ and θ are dependent. Why is this dependence useful?

5. The prior distribution for θ has a prior standard deviation; let's denote it σ. How might we interpret σ?

6. The prior standard deviation σ is also a parameter, so we could place a prior distribution on it. What would the prior distribution on σ represent?

- A Bayesian hierarchical model consists of the following layers in the hierarchy.
  1. A model for the data y which depends on model parameters θ and is represented by the likelihood $f(y|\theta)$.
     - The model usually involves multiple parameters. Generically, θ represents a vector of parameters that show up in the likelihood function.
     - Remember: While the likelihood for a particular y sample is computed by evaulating the conditional density of y given θ, the likelihood function is *a function of θ, treating the data y as fixed.*
  2. A (joint) prior distribution for the model parameter θ, denoted $\pi(\theta|\phi)$, which depends on *hyperparameters* ϕ.
  3. A *hyperprior* distribution, i.e., a prior distribution for the hyperparameters, denoted $\pi(\phi)$. (Again, ϕ generically represents a vector of hyperparameters.)
- The above specification implicitly assumes that the data distribution and likelihood function only depend on θ; the hyperparameters only affect y through θ. That is, the general hierarchical model above assumes that the data y and hyperparameters ϕ are conditionally independent given the model parameters θ, so the likelihood satisfies

  $$f(y|\theta, \phi) = f(y|\theta)$$

  The joint prior distribution on parameters and hyperparameters is

  $$\pi(\theta, \phi) = \pi(\theta|\phi)\pi(\phi)$$

- Given the observed data y, we find the joint posterior density of the parameters and hyperparameters, denoted $\pi(\theta, \phi|y)$, using Bayes rule: posterior $\propto$ likelihood $\times$ prior.

$$\begin{aligned}\pi(\theta, \phi|y) &\propto f(y|\theta, \phi)\pi(\theta, \phi) \\ &\propto f(y|\theta)\pi(\theta|\phi)\pi(\phi)\end{aligned}$$

- In Bayesian hierarchical models, there will generally be dependence between parameters. But dependence among parameters is actually useful for several reasons:
  - The dependence is natural and meaningful in the problem context.
  - If parameters are dependent, then all data can inform parameter estimates.

Example 24.3. Now suppose want to estimate the free throw percentages for J NBA players, given data consisting of the number of free throws attempted n_j and made y_j by player j. Specify a hierarchical model, including your choice of prior distribution. How many parameters are there in the model?

Example 24.4. Use software to fit the Bayesian model using the 76ers data. Summarize the posterior distribution of μ.

Example 24.5. For each player j, consider the posterior mean and posterior SD of θ_j.

1. What is the posterior mean of θ_j for players with no attempts? Why does this make sense?

2. Why is the posterior mean of θ_j not the same for players with no attempts?

3. For which players is the posterior mean closest to the observed free throw percentage? Why?

4. Which players have the smallest posterior SDs? Why?

5. Find and interpret 50%, 80%, and 98% posterior credible intervals for Joel Embiid.

6. Find and interpret 50%, 80%, and 98% posterior credible intervals for Kevin Ross.

25 Introduction to Bayesian Hierarchical Linear Models

We have studied a Bayesian simple linear regression model. We have also introduced some basic principles of hierarchical models. Now we'll combine elements to formulate Bayesian hierarchical linear regression models.

This handout corresponds to Chapters 15-17 of (Johnson, Ott, and Dogucu 2022). This handout only contains a brief introduction; see (Johnson, Ott, and Dogucu 2022) for a much more thorough analysis.

We'll consider the `cherry_blossom_sample` data in the `bayesrules` package which contains the `net` running times (in minutes) and `age` for participants in the annual 10-mile Cherry Blossom race held in Washington, D.C. Each runner in the sample is in their 50s or 60s.

```
library(bayesrules)

data(cherry_blossom_sample)
running <- cherry_blossom_sample |>
  select(runner, age, net) |>
  na.omit()

head(running)
```

```
# A tibble: 6 x 3
  runner   age   net
  <fct>  <int> <dbl>
1 1         53  84.0
2 1         54  74.3
3 1         55  75.2
4 1         56  74.2
5 1         57  74.2
6 2         52  82.7
```

1. Suppose we fit a Bayesian simple regression model of `net` on `age`. What do the slope and intercept parameters represent in context? What might be a reasonable prior?

(You can assume that the `age` variable has been centered by subtracting the sample mean.)

2. Fit the model from the previous part (see the results in Section 17.1 of (Johnson, Ott, and Dogucu 2022)). Do the results seem reasonable? (Hint: consider the posterior distribution of the slope.)

3. Each of the runners in the sample has run the race in multiple years. What assumption of the simple linear regression model is violated?

4. Formulate a Bayesian hierarchical linear model which assumes the intercept (but not slope) varies by runner. What do the parameters represent in context?

5. Fit the model from the previous part (see the results in Section 17.2.4 of (Johnson, Ott, and Dogucu 2022)). What are some features of the analysis you would explore? Consider the posterior distribution of the slope associated with `age`; does the posterior distribution based on this hierarchical model seem more reasonable than the one based on the simple linear regression model?

6. Formulate a Bayesian hierarchical linear model which assumes both the intercept and the slope vary by runner. What do the parameters represent in context?

7. Fit the model from the previous part (see the results in Section 17.3.3 of (Johnson, Ott, and Dogucu 2022)). What are some features of the analysis you would explore?

8. Compare the two hierarchical regression models. What are some similarities? What are some differences?

26 Naive Bayes Classification

We have seen how to fit Bayesian regression models for predicting numerical variables. Now we will introduce a Bayesian *classification* method for predicting *categorical* variables. The goal is to "classify" observations according to which category (class) they belong to.

This handout corresponds to Chapter 14 of (Johnson, Ott, and Dogucu 2022).

There exist multiple penguin species throughout Antarctica, including the *Adelie* (A), *Chinstrap* (C), and *Gentoo* (G). Our goal will be to classify penguins according to their species given characteristics like weight or bill length. (Throughout "penguin" refers to an Antarctic penguin of one of these three species.)

Example 26.1. Suppose first that we have no information about a randomly selected penguin (other than it's an Antarctic penguin from one of these three species).

1. How might we formulate a prior distribution for the species of the penguin? Would we necessarily assume equal prior probability for the three species?

2. Suppose that among Antarctic penguins of these three species, 44.2% of Adelie, 19.8% are Chinstrap, and 36.0% are Gentoo. Start to set up a Bayes table.

Example 26.2. Now suppose we know that a randomly selected penguin is below average weight (that is, below 4200g).

1. What is the next column of the Bayes table to fill in? What information would we need to do so?

2. Suppose that 83.4% of Adelie penguins are below average weight, 89.7% of Chinstrap penguins are below average weight, and 4.9% of Gentoo penguins are below average weight. Among these three species, the likelihood of being below average weight is greatest for Chinstrap. Does that mean we should necessarily classify the penguin to be Chinstrap? Why?

3. Complete the Bayes table and find the posterior probability of each species given that the penguin is below average weight.

4. If the species is below average weight, what species would you classify it as? Why?

5. Now suppose the the species is not below average weight. Compute the posterior probability of each species given that the penguin is not below average weight. What species would you classify it as? Why?

6. Suppose that you classify any randomly selected penguin based on whether or not it is below average weight. What is the posterior probability that you classify the penguin

correctly?

Example 26.3. Now suppose we know that a randomly selected penguin has a bill length of 50mm. We'll start by assuming we don't know yet whether or not it is below average weight.

1. Start to create a Bayes table. What is the prior? What is the next column of the Bayes table to fill in? What information would we need to do so?

2. Suppose that bill lengths (mm) follow a N(38.8, 2.66) distribution for Adelie, a N(48.8, 3.34) distribution for Chinstrap, and a N(47.5, 3.08) distribution for Gentoo. How would you fill in the likelihood column?

3. Among these three species, the likelihood of having of a 50mm bill is greatest for Chinstrap. Does that mean we should necessarily classify the penguin to be Chinstrap? Why?

4. Complete the Bayes table and find the posterior probability of each species given that the penguin has a 50mm bill.

5. If the species has a 50mm bill, what species would you classify it as? Why?

6. Now suppose before measuring bill length we had already known that the penguin was below average weight. How would our Bayes table have changed? If we don't change the likelihood column, what are we assuming?

7. Compute the posterior probability of each species given that the penguin is below average weight with a 50mm bill, assuming conditional independence between weight and bill length. Given a species with these characteristics, what species would you classify it as? Why?

- **Naive Bayes classification** uses Bayes rule to predict the class of a categorical response variable y given predictor variables $x_1, \ldots, x_p$
- Predictor variables can be categorical or numerical
- Naive Bayes classification assumes that predictors are *conditionally independent* given each class
- Naive Bayes classification typically assumes that any numerical predictor variable is conditionally Normal
- Given observed data
  - Prior probabilities for the response variable are estimated to be the observed sample proportion for each class
  - For a categorical predictor x, the likelihood of value $\tilde{x}$ is estimated to be the observed sample proportion of $\tilde{x}$ within each class
  - For a numerical predictor x, the likelihood is computed assuming that within each class values of the predictor follow a Normal distribution with mean and SD estimated to be the sample mean and SD within each class.

Example 26.4. Now suppose we know that a penguin has a bill length of 50mm and a flipper length of 195mm. (We'll ignore whether or not it is below average weight for this example.)

1. Starting with our original prior, what is the "evidence" that the likelihood is based on? What information do we need to fill in the likelihood column?

2. Suppose that flipper lengths (mm) follow a N(190, 6.54) distribution for Adelie, a N(196, 7.13) distribution for Chinstrap, and a N(217, 6.48) distribution for Gentoo. (Recall that bill lengths (mm) follow a N(38.8, 2.66) distribution for Adelie, a N(48.8, 3.34) distribution for Chinstrap, and a N(47.5, 3.08) distribution for Gentoo.) How would you fill in the likelihood column? What are we assuming?

3. Complete the Bayes table and find the posterior probability of each species given that the penguin has a 50mm bill and a 195mm flipper.

4. If the species has a 50mm bill and a 195mm flipper, what species would you classify it as? Why?

27 Introduction to Bayesian Model Comparison

- A Bayesian model is composed of both a model for the data (likelihood) and a prior distribution on model parameters
- **Model selection** usually refers to comparing different models[1] for the data (likelihoods); for example, selecting which predictors to include in a regression or classification model, selecting between Normal or t error distribution, comparing a hierarchical model to a pooled model, etc.
- But model selection can also concern choosing between models with the same likelihood but different priors.
- In Bayesian model comparison, prior probabilities are assigned to each of the models, and these probabilities are updated given the data according to Bayes rule.

27.1 Odds and Bayes factors

- The **odds** of an event is a ratio involving the probability that the event occurs and the probability that the event does not occur

$$\text{odds}(A) = \frac{P(A)}{P(A^c)} = \frac{P(A)}{1 - P(A)}$$

- In many situations (e.g., gambling) odds are reported as odds *against* A, that is, the odds of A^c: $P(A^c)/P(A)$.
- The probability of an event can be obtained from odds

$$P(A) = \frac{\text{odds}(A)}{1 + \text{odds}(A)}$$

Example 27.1. The ELISA test for HIV was widely used in the mid-1990s for screening blood donations. As with most medical diagnostic tests, the ELISA test is not perfect. If a person actually carries the HIV virus, experts estimate that this test gives a positive result 97.7% of

[1] Tools that are commonly used in frequentist analysis — such as cross-validation — can be applied to compare Bayesian models. However, we're going to focus on some specifically Bayesian model comparison ideas.

the time. (This number is called the *sensitivity* of the test.) If a person does not carry the HIV virus, ELISA gives a negative (correct) result 92.6% of the time (the *specificity* of the test). Estimates at the time were that 0.5% of the American public carried the HIV virus (the *base rate*).

1. Calculate the *prior odds* of a randomly selected American having the HIV virus, before taking an ELISA test.

2. Suppose that a randomly selected American tests positive. Compute the posterior probability that the person actually carries the virus. (Why is this probability so small?)

3. Calculate the *posterior odds* of a randomly selected American having the HIV virus, given a positive test result.

4. By what factor has the *odds* of carrying HIV increased, given a positive test result, as compared to before the test? This is called the **Bayes factor**.

5. Suppose you were given the prior odds and the Bayes factor. How could you compute the posterior odds?

6. Compute the ratio of the likelihoods of testing positive, for those who carry HIV and for those who do not carry HIV. What do you notice?

- The **Bayes factor (BF)** in favor of hypothesis H given evidence E is defined to be the ratio of the posterior odds to the prior odds

$$BF = \frac{\text{posterior odds}}{\text{prior odds}} = \frac{P(H|E)/P(H^c|E)}{P(H)/P(H^c)}$$

- The Bayes factor in favor of hypothesis H given evidence E can be calculated as the *ratio of the likelihoods*

$$BF = \frac{P(E|H)}{P(E|H^c)}$$

- **Odds form of Bayes rule**

$$\frac{P(H|E)}{P(H^c|E)} = \frac{P(H)}{P(H^c)} \times \frac{P(E|H)}{P(E|H^c)}$$
$$\text{posterior odds} = \text{prior odds} \times \text{Bayes factor}$$

27.2 Posterior probability of models

Example 27.2. Suppose I have some trick coins[2], some of which are biased in favor of landing on heads, and some of which are biased in favor of landing on tails. I will select a trick coin at random; let θ be the probability that the selected coin lands on heads in any single flip. I will flip the coin n times and use the data to decide about the direction of its bias. This can be viewed as a choice between two models

- Model 1: the coin is biased in favor of landing on heads
- Model 2: the coin is biased in favor of landing on tails

1. Assume that in model 1 the prior distribution for θ is Beta(7.5, 2.5). Suppose in $n = 10$ flips there are 6 heads. Use simulation to approximate the probability of observing 6 heads in 10 flips *given that model 1 is correct*.

[2]The coin flipping examples in this section are adapted from Chapter 10 of (*Doing Bayesian Data Analysis* 2015).

2. Assume that in model 2 the prior distribution for θ is Beta(2.5, 7.5). Suppose in $n = 10$ flips there are 6 heads. Use simulation to approximate the probability of observing 6 heads in 10 flips *given that model 2 is correct*.

3. Suppose I flip the coin 10 times and 6 of the flips land on heads. Use the simulation results to approximate and interpret the Bayes Factor in favor of model 1 given 6 heads in 10 flips.

4. Suppose our prior probability for each model was 0.5. Find the posterior probability of each model given 6 heads in 10 flips.

5. Suppose I know I have a lot more tail biased coins, so my prior probability for model 1 was 0.1. Find the posterior probability of each model given 6 heads in 10 flips.

27.3 Model averaging

Example 27.3. Continuing Example 27.2. Now suppose I want to predict the number of heads in the next 10 flips of the selected coin, given that there were 6 heads in the first 10 flips.

1. Use simulation to approximate the posterior predictive distribution of the number of heads in the next 10 flips given 6 heads in the first 10 flips *given that model 1 is the correct model*. In particular, approximate the posterior predictive probability that there are 7 heads in the next 10 flips *given then model 1 is the correct model*.

2. Use simulation to approximate the posterior predictive distribution of the number of heads in the next 10 flips given 6 heads in the first 10 flips *given that model 2 is the correct model*. In particular, approximate the posterior predictive probability that there are 7 heads in the next 10 flips *given then model 2 is the correct model*.

3. Suppose our prior probability for each model was 0.5. Use simulation to approximate the posterior predictive distribution of the number of heads in the next 10 flips given 6 heads in the first 10 flips. In particular, approximate the posterior predictive probability that there are 7 heads in the next 10 flips.

- Bayesian model comparison can be viewed as Bayesian estimation in a hierarchical model with an extra level for "model".
- In Bayesian model comparison, prior probabilities are assigned to each of the models, and these probabilities are updated given the data according to Bayes rule.
- When several models are under consideration, the Bayesian model is the full hierarchical structure which spans all models being compared.
- Thus, the most complete posterior prediction takes into account all models, weighted by their posterior probabilities.

- That is, prediction is accomplished by taking a weighted average across the models, with weights equal to the posterior probabilities of the models. This is called **model averaging**.

27.4 Accounting for model complexity

Example 27.4. Suppose again I select a coin, with probability of heads θ, but now the decision is whether the coin is fair. Suppose we consider the two models

- "Must be fair" model: prior distribution for θ is Beta(500, 500)
- "Anything is possible" model: prior distribution for θ is Beta(1, 1)

1. Suppose we observe 15 heads in 20 flips. Use simulation to approximate the Bayes factor in favor of the "must be fair" model given 15 heads in 20 flips. Which model does the Bayes factor favor?

2. Suppose we observe 11 heads in 20 flips. Use simulation to approximate the Bayes factor in favor of the "must be fair" model given 11 heads in 20 flips. Which model does the Bayes factor favor?

3. The "anything is possible" model has any value available to it, including 0.5 and the sample proportion 0.55. Why then is the "must be fair" option favored in the previous part?

- Complex models generally have an inherent advantage over simpler models because complex models have many more options available, and one of those options is likely to fit the data better than any of the fewer options in the simpler model.
- However, we don't always want to just choose the more complex model. Always choosing the more complex model overfits the data.
- Bayesian model comparison naturally compensates for discrepancies in model complexity.
- In more complex models, prior probabilities are diluted over the many options available.
- The numerator and denominator in the Bayes factor are *average* likelihoods: the likelihood of the data averaged over each possible value of θ.
- Even if a complex model has some particular combination of parameters that fit the data well, the prior probability of that particular combination is likely to be small because the prior is spread more thinly than for a simpler model, which pulls down the average likelihood of the complex model.
- Thus, in Bayesian model comparison, a simpler model can "win" if the data are consistent with it, even if the complex model fits well.

27.5 Sensitivity of model comparison to prior

Example 27.5. Continuing Example 27.5 where we considered the two models

- "Must be fair" model: prior distribution for θ is Beta(500, 500)
- "Anything is possible" model: prior distribution for θ is Beta(1, 1)

Suppose we observe 65 heads in 100 flips.

1. Use simulation to approximate the Bayes factor in favor of the "must be fair" model given 65 heads in 100 flips. Which model does the Bayes factor favor?

2. We have discussed different notions of a "non-informative/vague" prior. We often think of Beta(1, 1) = Uniform(0, 1) as a non-informative prior, but there are other considerations. In particular, a Beta(0.01, 0.01) is often used a non-informative prior in this context. Think of a Beta(0.01, 0.01) prior like an approximation to the improper Beta(0, 0) prior based on "no prior successes or failures". Suppose now that the "anything is possible" model corresponds to a Beta(0.01, 0.01) prior distribution for θ. Use simulation to approximate the Bayes factor in favor of the "must be fair" model given 65 heads in 100 flips. Which model does the Bayes factor favor?

3. Is the *choice of model* — that is, which model the Bayes factor favors — sensitive to the change of prior distribution within the "anything is possible" model?

4. For each of the two "anything is possible" priors, find the posterior distribution of θ and a 98% posterior credible interval for θ given 65 heads in 100 flips. Is *estimation of* θ within the "anything is possible" model sensitive to the change in the prior distribution for θ?

- In Bayesian *estimation of continuous parameters within a model*, the posterior distribution is typically not too sensitive to changes in prior (provided that there is a reasonable amount of data and the prior is not too strict).
- In contrast, in Bayesian *model comparison*, the posterior probabilities of the models and the Bayes factors can be extremely sensitive to the choice of prior distribution within each model.
- When comparing different models, prior distributions on parameters within each model should be equally informed. One strategy is to use a small set of "training data" to inform the prior of each model before comparing.

Example 27.6. Continuing Example 27.5 where we considered two priors in the "anything is possible model": Beta(1, 1) and Beta(0.1, 0.1). We will again compare the "anything is possible model" to the "must be fair" model which corresponds to a Beta(500, 500) prior.

Suppose we observe 65 heads in 100 flips.

1. Assume the "anything is possible" model corresponds to the Beta(1, 1) prior. Suppose that in the first 10 flips there were 6 heads. Compute the posterior distribution of θ in each of the models *after the first 10 flips*. Then use simulation to approximate the

Bayes factor in favor of the "must be fair" model given 59 heads in the remaining 90 flips, using the posterior distribution of θ after the first 10 flips as the prior distribution in the simulation. Which model does the Bayes factor favor?

2. Repeat the previous part assuming the "anything is possible" model corresponds to the Beta(0.01, 0.01) prior. Compare with the previous part.

27.6 Hypothesis testing (just don't do it)

Example 27.7. Consider a null hypothesis significance test of $H_0 : \theta = 0.5$ versus $H_1 : \theta \neq 0.5$. How does this situation resemble the previous problems in this section?

- A null hypothesis significance test can be viewed as a problem of Bayesian model selection in which one model has a prior distribution that places all its credibility on the null hypothesized value. However, is it really plausible that the parameter is *exactly* equal to the hypothesized value?
- Unfortunately, this model-comparison (Bayes factor) approach to testing can be extremely sensitive to the choice of prior corresponding to the alternative hypothesis.
- An alternative Bayesian approach to testing involves choosing a **region of practical equivalence (ROPE).** A ROPE indicates a range of parameter values that are considered to be "practically equivalent" to the null hypothesized value.
  - A credible interval for the parameter can be compared to the ROPE to see if they overlap.

  - How do you choose the ROPE? That determines on the practical application.

- In general, traditional testing of point null hypotheses (that is, "*no* effect/difference") is not a primary concern in Bayesian statistics.
- Rather, the *posterior distribution* provides all relevant information to make decisions about practically meaningful issues. Ask research questions that are important in the context of the problem and use the posterior distribution to answer them.

Example 27.8. Recall a previous application: Does uniform color give athletes an advantage over their competitors? To investigate this question, Hill and Barton (*Nature*, 2005) examined the records in the 2004 Olympic Games for four combat sports: boxing, tae kwon do, Greco-Roman wrestling, and freestyle wrestling. Competitors in these sports were randomly assigned to wear either a red or a blue uniform. Let the parameter θ represent the probability that a competitor wearing *red* wins a match.

In this context, we might say that if the probability of winning by the competitor wearing red is between 0.49 and 0.51, then any color advantage is small enough to be considered unimportant. That is, we might consider the interval [0.49, 0.51] as a ROPE for 0.5. Therefore, we consider a null hypothesis of $H_0 : 0.49 \leq \theta \leq 0.51$ and an alternative that θ is outside of [0.49, 0.51].

The researchers collected data on a sample of 457 matches; the competitor wearing red won in 248 of the matches.

1. Assume a Uniform(0, 1) = Beta(1, 1) prior distribution for θ. Compute the prior odds in favor of the alternative hypothesis, the posterior odds in favor of the alternative hypothesis, and the Bayes Factor in favor of the alternative hypothesis given the sample data. (You can compute probabilities using `pbeta`; you don't need simulation.) Which hypothesis does the Bayes factor favor?

2. Assume a Beta(10, 10) prior distribution for θ. Compute the prior odds in favor of the alternative hypothesis, the posterior odds in favor of the alternative hypothesis, and the Bayes Factor in favor of the alternative hypothesis given the sample data. (You can compute probabilities using `pbeta`; you don't need simulation.) Which hypothesis does the Bayes factor favor?

3. Assume a Beta(200, 200) prior distribution for θ. Compute the prior odds in favor of the alternative hypothesis, the posterior odds in favor of the alternative hypothesis, and the Bayes Factor in favor of the alternative hypothesis given the sample data. (You can compute probabilities using `pbeta`; you don't need simulation.) Which hypothesis does the Bayes factor favor?

4. Is the Bayes Factor sensitive to the choice of prior?

5. Consider each of the three prior distributions in this example. Which of the three prior distribution places the highest prior probability on the alternative hypothesis being true? Which of the three prior distribution leads to the largest Bayes factor in favor of the alternative hypothesis? Can you explain why?

Example 27.9. Continuing Example 27.9. An alternative to hypothesis testing is to just use credible intervals to determine ranges of plausible values for the parameters, and to assess any hypotheses or regions of practical equivalence in light of the credible values.

1. Assume a Uniform(0, 1) = Beta(1, 1) prior for θ. Compute a posterior 98% credible interval for θ given the sample data.

hypothesis	prior	likelihood	product	posterior
Carries HIV	0.005	0.9770	0.0049	0.0622
Does not carry HIV	0.995	0.0740	0.0736	0.9378
sum	1.000	NA	0.0785	1.0000
ratio	0.005	13.2027	0.0663	0.0663

2. Assume a Beta(10, 10) prior for θ. Compute a posterior 98% credible interval for θ given the sample data.

3. Assume a Beta(200, 200) prior for θ. Compute a posterior 98% credible interval for θ given the sample data.

4. Are the posterior 98% credible intervals highly sensitive to the choice of prior?

5. Assuming this is a representative sample, does the data provide evidence that θ lies outside of [0.49, 0.51]?